report_32bit

April 12, 2025

Slides and markers

$AS1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$AS2_Les [1] CD11c CD123 CD279 CD3 CD4 CD4_raw CD8

$AS3_Les [1] CD279 CD3 CD4 CD56 CD8

$CF_8wph character(0)

$CF1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD8

$CF2_Les [1] CD11c CD123 CD279 CD3 CD4 CD8

$CF5_Les [1] CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$CJ1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$CJ2_Les [1] CD11c CD123 CD15 CD3 CD4 CD45 CD68 CD8

$DD_8wph character(0)

$DD1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$DD2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$DD3_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$EB_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD8

$JK_8wph character(0)

$JK1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD56 CD68
[9] CD68_raw CD69 CD8

$JK2_Les [1] CD3 CD4 CD8

$JK3_Les [1] CD123 CD15 CD279 CD3 CD4 CD68 CD68_raw CD69
[9] CD8

$JM1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$JM2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$JM3_Les [1] CD3 CD8

$JT_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD8

$LD_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$LG1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD69 CD8

$LG2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$LG3_Les [1] CD3 CD4 CD68 CD68_raw CD8

$LM_9wph character(0)

$LM_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$LS_8wph character(0)

$LS1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$LS2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$LS3_Les [1] CD3 CD4 CD68 CD68_raw CD8

$SK1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD8

$SK2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$SK3_8wph_in_use character(0)

$SK3_Les [1] CD11c CD123 CD279 CD3 CD4 CD68 CD68_raw CD8

There are 13 lesion biopsies in total, including AS. Different biopsy samples or even different slices from the same biopsy sample may have different amount of lesion tissues. Some completely lack lesion.

The key markers are: CD CD11c CD123 CD15 CD56 CD68. Note that the CD68 antibody stained the epidermis of some tissues. CD68_raw.png is the raw image, and CD68.png is the processed image with reduced epidermis brightness to improve quality of segmentation. Furthermore, for CF lesion CD68 staining, CF5_Les instead of CF1_Les is used due to quality reasons.

Many lesion biopsies have multiple sections. The following choices of representative slides are made:

- AS1_Les
- CF1_Les
- CJ1_Les
- DD1_Les
- EB_Les
- JK1_Les
- JM1_Les
- JT_Les
- LD_Les
- LG2_Les
- LM_Les
- LS2_Les
- SK2_Les

Scatterplots of Shedding rate and counts and densities

In all plots, R2* is computed without slides from AS_Les. Two versions of the plots are shown: one with all slides and one with only the selected slides.

First, we count cells in regions regularized by distance from DEJ

- DEJ500: dermis up to 500 um below DEJ
- DEJ1000: dermis up to 1000 um below DEJ
- DEJInf: all dermis

Regularized densities by regions. DI: dermis intact; DU: dermis ulcerative; EI: epidermis intact; EU: epidermis ulcerative. SR: shedding rate.

Table 1: DEJ500

	SR	EI	DI	EU	DU	initial	total
AS1_Les	83.9	959	16568	0	0	AS	17527
CF1_Les	17.5	1787	22743	2077	28866	CF	55473
CJ1_Les	43.1	539	4661	0	0	CJ	5200
DD1_Les	39.4	1741	24708	489	13512	DD	40450
EB_Les	4.6	575	25489	0	0	EB	26064
JK1_Les	23.4	50	985	0	0	JK	1035
JM1_Les	24.6	1067	23153	916	21119	JM	46255
JT_Les	54.3	410	3629	0	0	JT	4039
LD_Les	18.8	2094	18412	0	0	LD	20506
LG2_Les	32.8	178	7872	0	0	LG	8050
LM_Les	4.4	646	21252	0	0	LM	21898
LS2_Les	31.0	1687	15380	0	0	LS	17067
SK2_Les	53.0	1266	11207	0	0	SK	12473

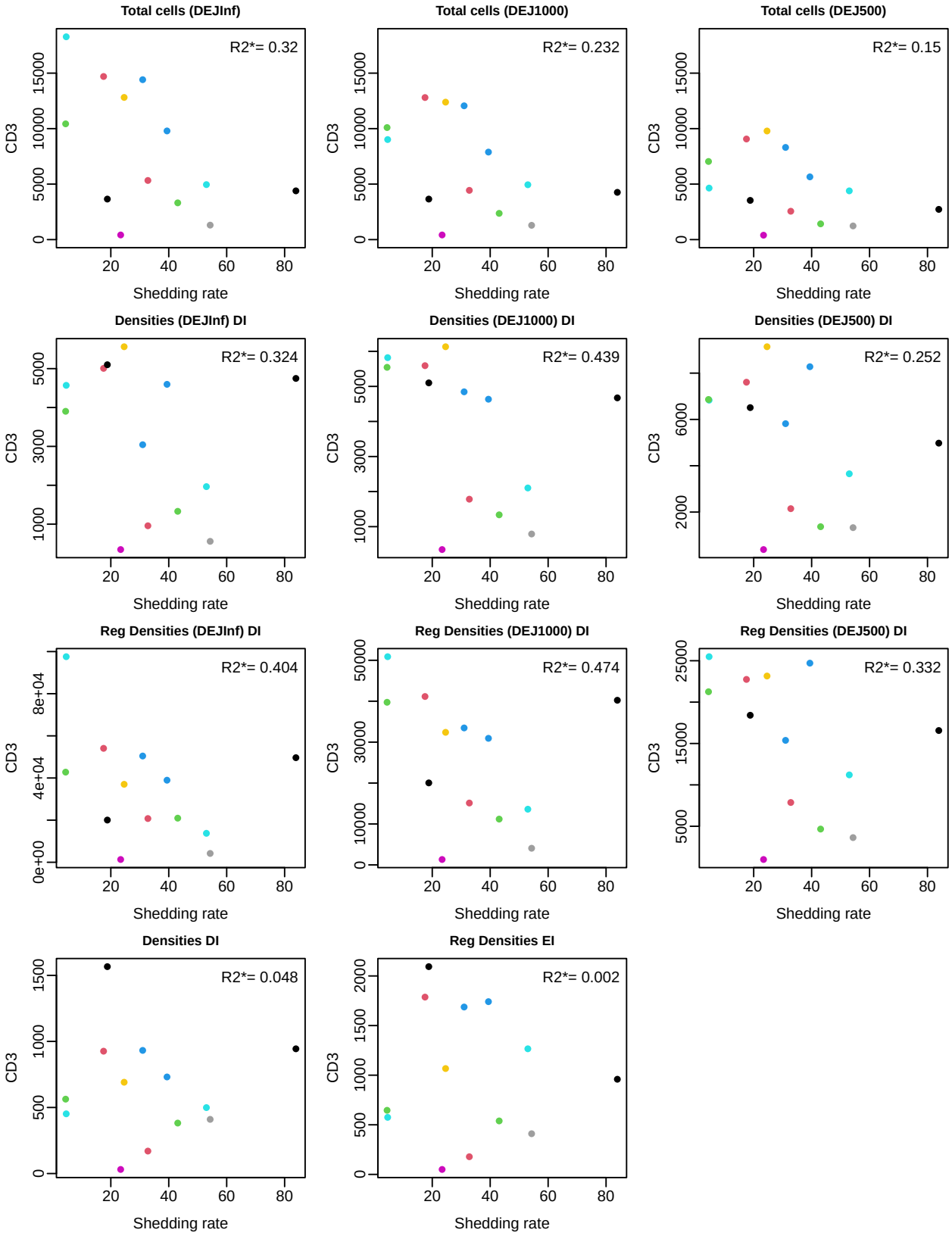
Table 2: DEJ1000

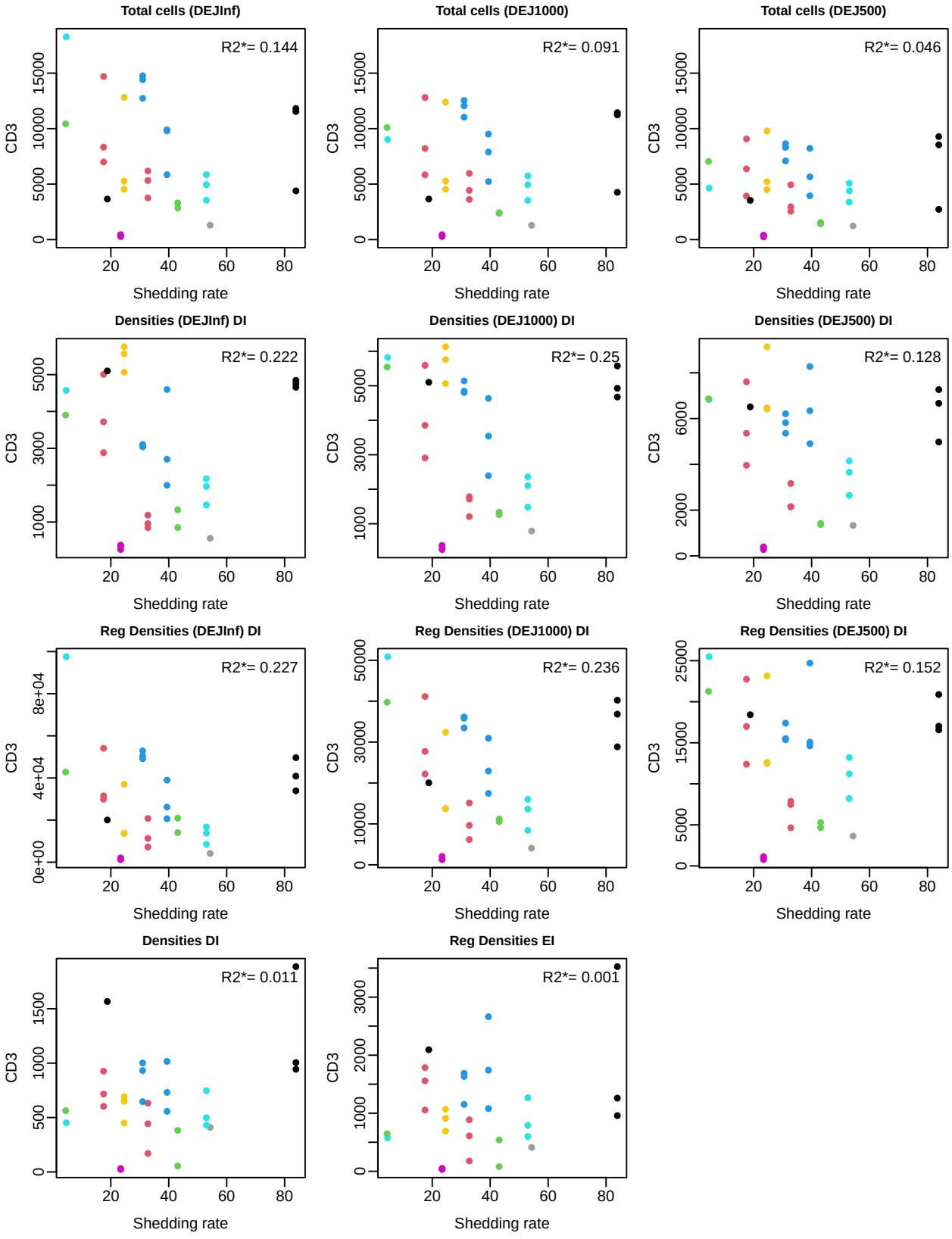
	SR	EI	DI	EU	DU	initial	total
AS1_Les	83.9	959	40233	0	0	AS	41192
CF1_Les	17.5	1787	41150	2077	48287	CF	93301
CJ1_Les	43.1	539	11192	0	0	CJ	11731
DD1_Les	39.4	1741	30930	489	21917	DD	55077
EB_Les	4.6	575	50879	0	0	EB	51454
JK1_Les	23.4	50	1321	0	0	JK	1371
JM1_Les	24.6	1067	32393	916	43238	JM	77614
JT_Les	54.3	410	4067	0	0	JT	4477
LD_Les	18.8	2094	20040	0	0	LD	22134
LG2_Les	32.8	178	15120	0	0	LG	15298
LM_Les	4.4	646	39738	0	0	LM	40384
LS2_Les	31.0	1687	33467	0	0	LS	35154
SK2_Les	53.0	1266	13609	0	0	SK	14875

Table 3: DEJInf

	SR	EI	DI	EU	DU	initial	total
AS1_Les	83.9	959	49636	0	0	AS	50595

	SR	EI	DI	EU	DU	initial	total
CF1_Les	17.5	1787	54073	2077	60997	CF	118934
CJ1_Les	43.1	539	20936	0	0	CJ	21475
DD1_Les	39.4	1741	38952	489	32584	DD	73766
EB_Les	4.6	575	97575	0	0	EB	98150
JK1_Les	23.4	50	1321	0	0	JK	1371
JM1_Les	24.6	1067	37006	916	53780	JM	92769
JT_Les	54.3	410	4194	0	0	JT	4604
LD_Les	18.8	2094	20040	0	0	LD	22134
LG2_Les	32.8	178	20755	0	0	LG	20933
LM_Les	4.4	646	42780	0	0	LM	43426
LS2_Les	31.0	1687	50433	0	0	LS	52120
SK2_Les	53.0	1266	13745	0	0	SK	15011


Total cells (DEJInf)
R2*= 0.32
Total cells (DEJ1000)
R2*= 0.232
Total cells (DEJ500)
R2*= 0.15
Densities (DEJInf) DI
R2*= 0.324
Densities (DEJ1000) DI
R2*= 0.439
Densities (DEJ500) DI
R2*= 0.252
Reg Densities (DEJInf) DI
R2*= 0.404
Reg Densities (DEJ1000) DI
R2*= 0.474
Reg Densities (DEJ500) DI
R2*= 0.332
Densities DI
R2*= 0.048
Reg Densities EI
R2*= 0.002
CD3
Shedding rate


Total cells (DEJInf)
R2*= 0.144
Total cells (DEJ1000)
R2*= 0.091
Total cells (DEJ500)
R2*= 0.046
Densities (DEJInf) DI
R2*= 0.222
Densities (DEJ1000) DI
R2*= 0.25
Densities (DEJ500) DI
R2*= 0.128
Reg Densities (DEJInf) DI
R2*= 0.227
Reg Densities (DEJ1000) DI
R2*= 0.236
Reg Densities (DEJ500) DI
R2*= 0.152
Densities DI
R2*= 0.011
Reg Densities EI
R2*= 0.001
CD3
Shedding rate

Next, we count cells in regions regularized by distance from skin

- Skin1000: up to 1000 um below skin
- SkinInf: all depth

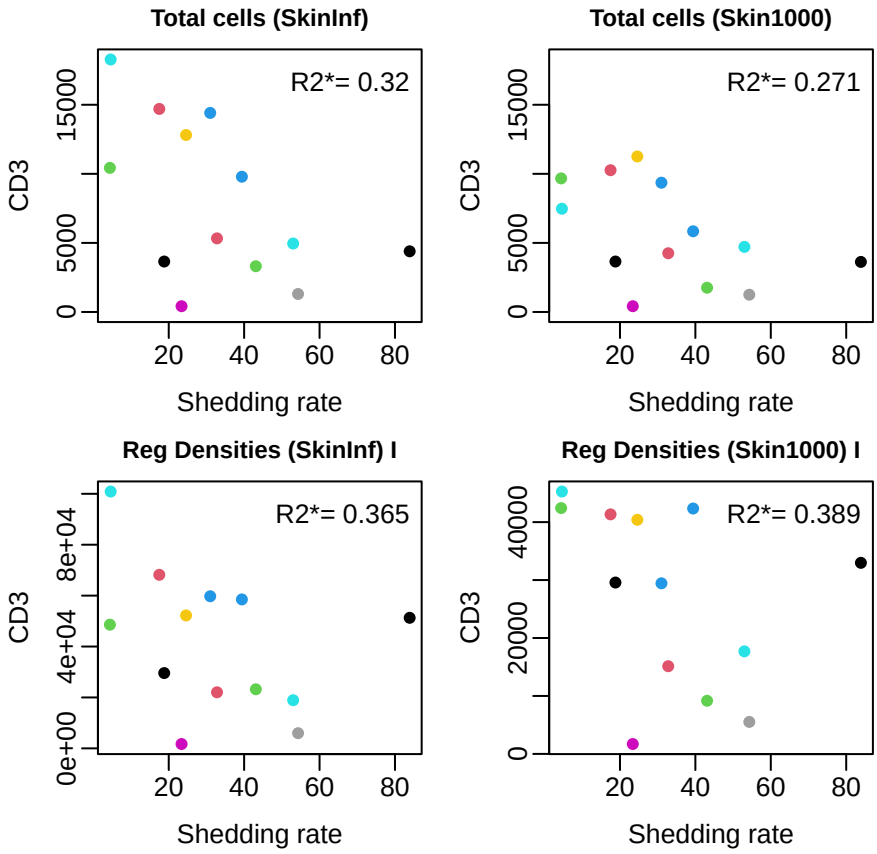
Regularized densities by regions. I: intact; U: ulcerative. SR: shedding rate.

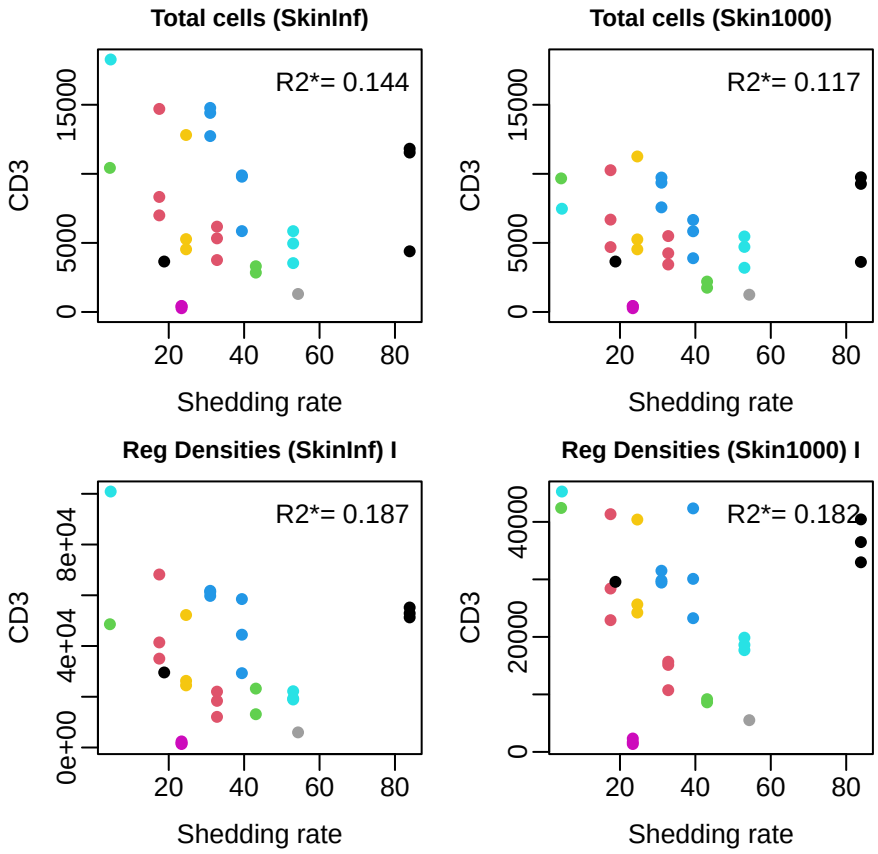
Table 4: Skin1000

	SR	I	U	initial	total
AS1_Les	83.9	32973	0	AS	32973
CF1_Les	17.5	41338	47917	CF	89255
CJ1_Les	43.1	9170	0	CJ	9170
DD1_Les	39.4	42345	17897	DD	60242
EB_Les	4.6	45277	0	EB	45277
JK1_Les	23.4	1708	0	JK	1708
JM1_Les	24.6	40402	43011	JM	83413
JT_Les	54.3	5519	0	JT	5519
LD_Les	18.8	29572	0	LD	29572
LG2_Les	32.8	15131	0	LG	15131
LM_Les	4.4	42423	0	LM	42423
LS2_Les	31.0	29439	0	LS	29439
SK2_Les	53.0	17696	0	SK	17696

Table 5: SkinInf

	SR	I	U	initial	total
AS1_Les	83.9	51284	0	AS	51284
CF1_Les	17.5	68180	75698	CF	143878
CJ1_Les	43.1	23215	0	CJ	23215
DD1_Les	39.4	58505	39893	DD	98398
EB_Les	4.6	100888	0	EB	100888
JK1_Les	23.4	1708	0	JK	1708
JM1_Les	24.6	52218	67348	JM	119566
JT_Les	54.3	5957	0	JT	5957
LD_Les	18.8	29572	0	LD	29572
LG2_Les	32.8	22021	0	LG	22021
LM_Les	4.4	48585	0	LM	48585
LS2_Les	31.0	59744	0	LS	59744
SK2_Les	53.0	18914	0	SK	18914


Total cells (SkinInf)
R2*= 0.32
CD3
0
5000
15000
20
40
60
80
Shedding rate
Total cells (Skin1000)
R2*= 0.271
CD3
0
5000
15000
20
40
60
80
Shedding rate
Reg Densities (SkinInf) I
R2*= 0.365
CD3
0e+00
4e+04
8e+04
20
40
60
80
Shedding rate
Reg Densities (Skin1000) I
R2*= 0.389
CD3
0
20000
40000
20
40
60
80
Shedding rate


Total cells (SkinInf)
R2*= 0.144
CD3
0
5000
15000
20
40
60
80
Shedding rate
Total cells (Skin1000)
R2*= 0.117
CD3
0
5000
15000
20
40
60
80
Shedding rate
Reg Densities (SkinInf) I
R2*= 0.187
CD3
0e+00
4e+04
8e+04
20
40
60
80
Shedding rate
Reg Densities (Skin1000) I
R2*= 0.182
CD3
0
20000
40000
20
40
60
80
Shedding rate

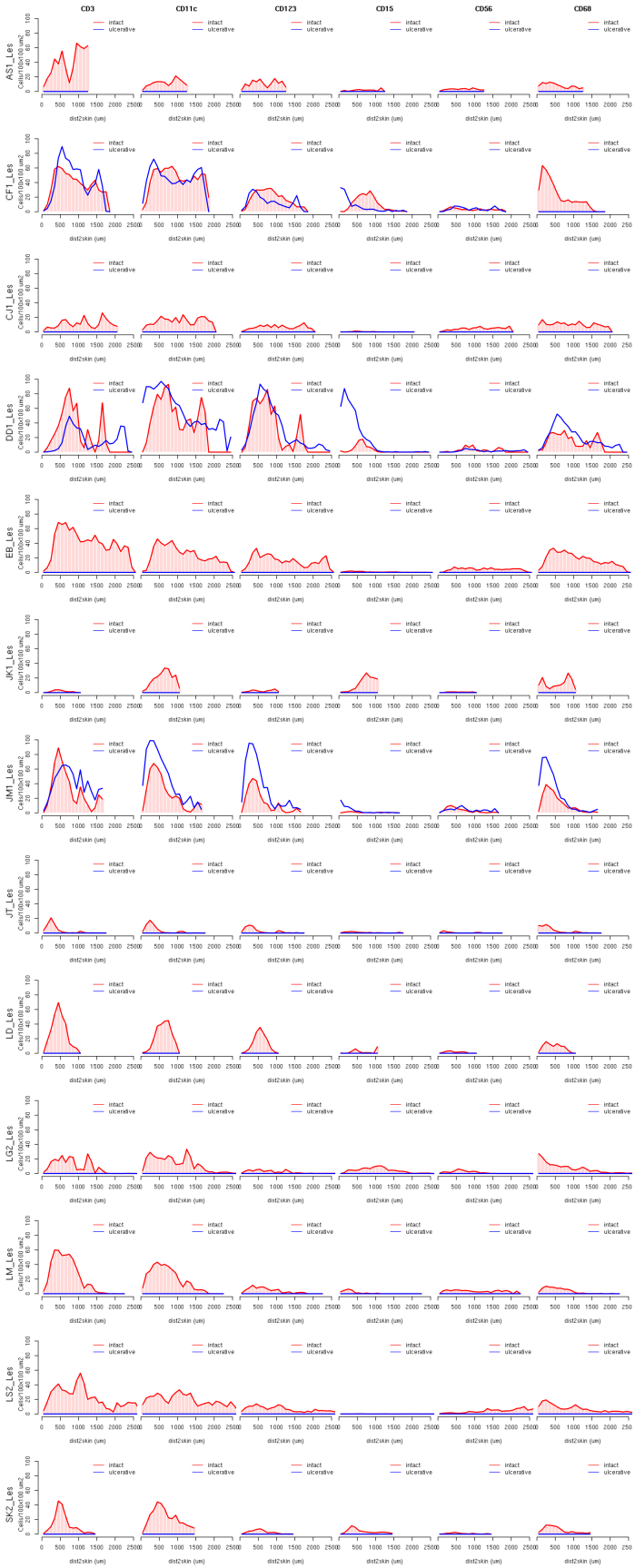
Density curves

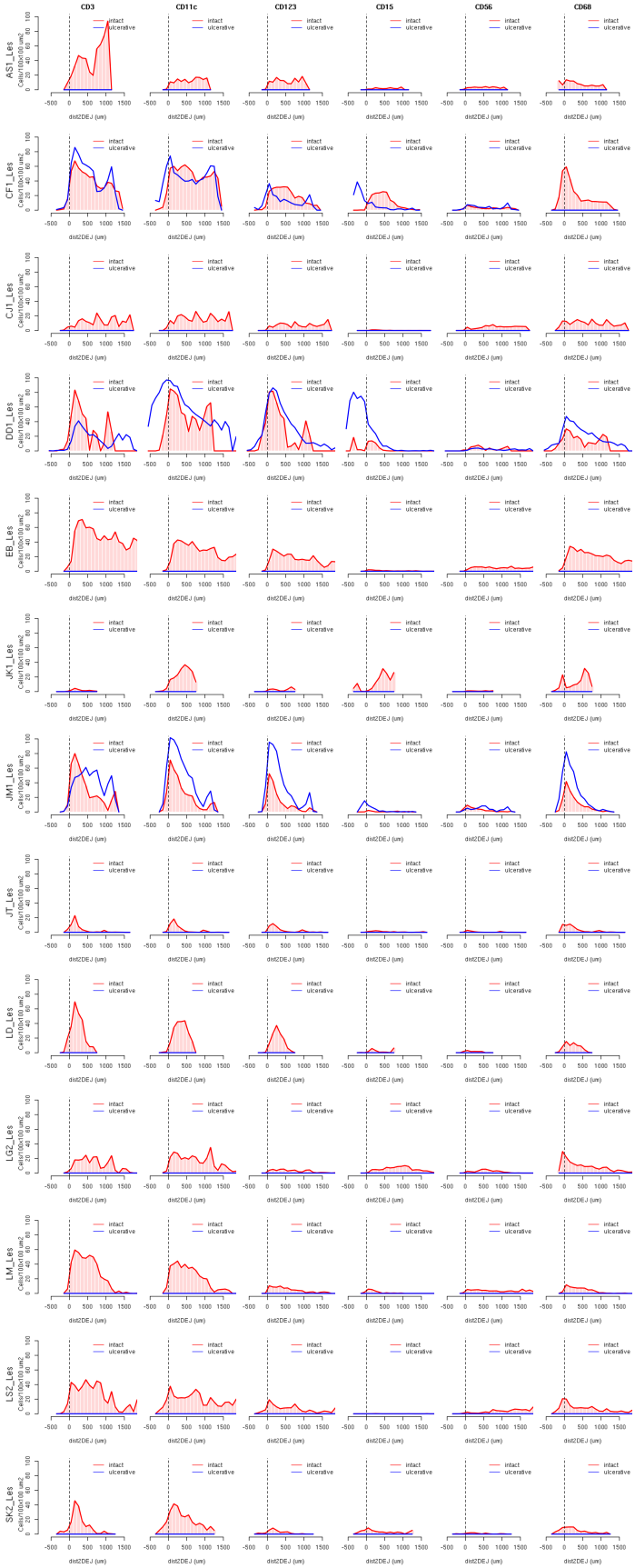
In this section we plot the number of cells per 100x100 um2 tissue as a function of distance to skin or DEJ. Each bin in the histogram represents the count in a 100x100 um2 area. Regularized density used in the correlates study is computed as the area under the curve with cutoff at a given depth.

Intact vs. ulcerrative - representative slides

In this subsection, there are two figures, one for dist2skin and one for dist2DEJ. Each figure shows 13 representative slides. Some observations:

- Comparing the CD3 cell distribution between CF and DD, we see that the ratio of the area under the curve across both epi and dermis between red and blue(ulcerative and intact) is about 1 for CD3/CF and 1/3 for CD3/DD. This suggests that CD3 is depleted in lesion in DD, but not in CF.
- Comparing the CD15 cell distribution in CF between intact and ulcerative, we see that the lesion peak is shifted to the left compared to the normal peak. This pattern is more clear in CF than in DD. We don't see this pattern for CD3.

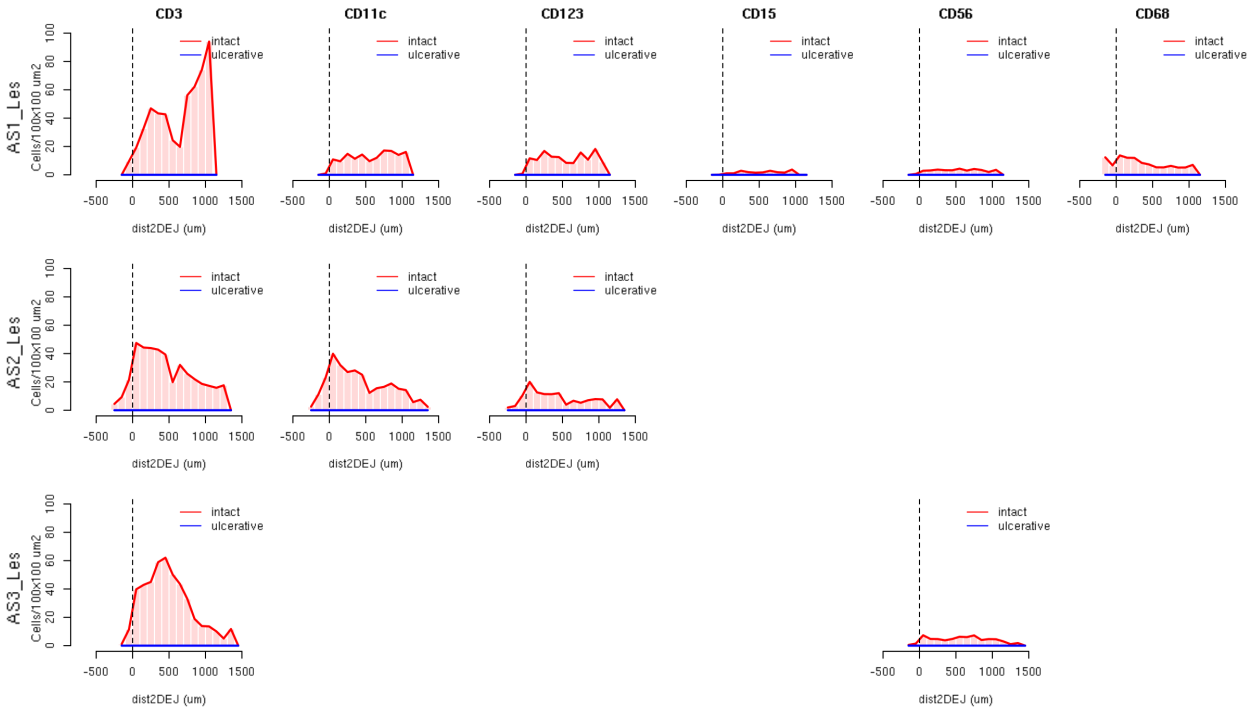

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
Cells/100x100 um2
dist2skin (um)

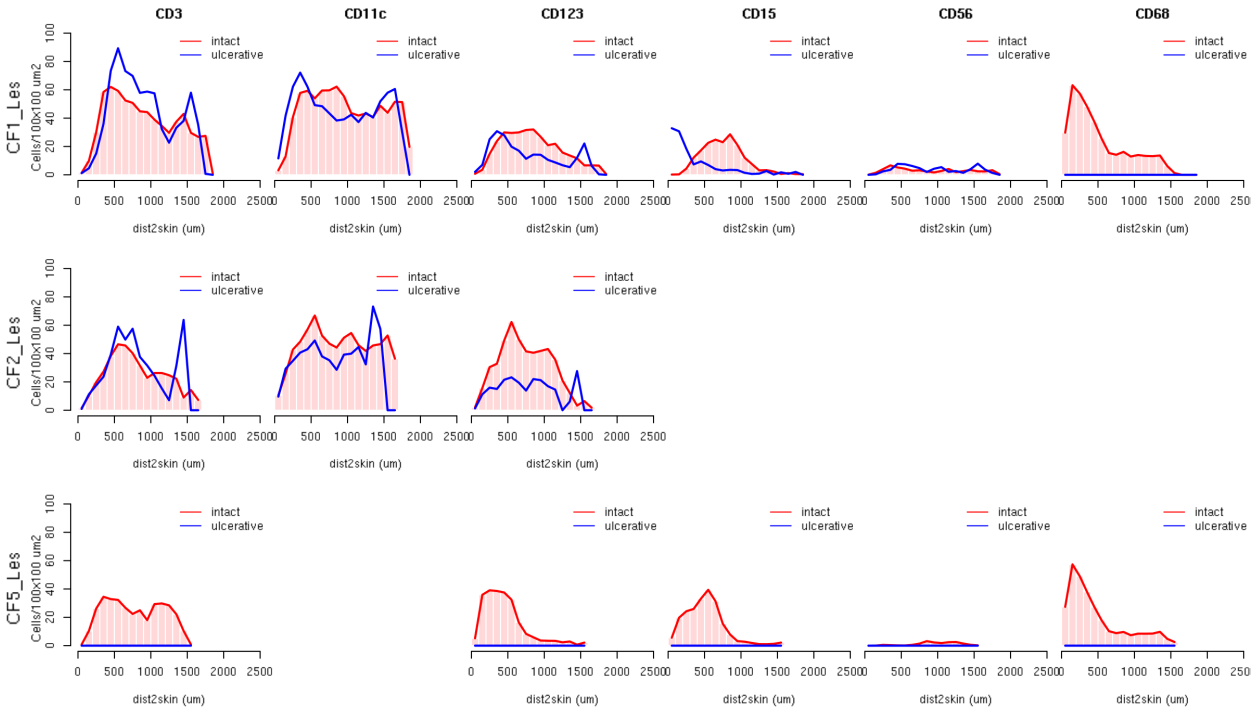

CD3
CD11c
CD123
CD15
CD56
CD68
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
Cells/100x100 um2
dist2DEJ (um)
intact
ulcerative

Intact vs. ulcerrative - all slides

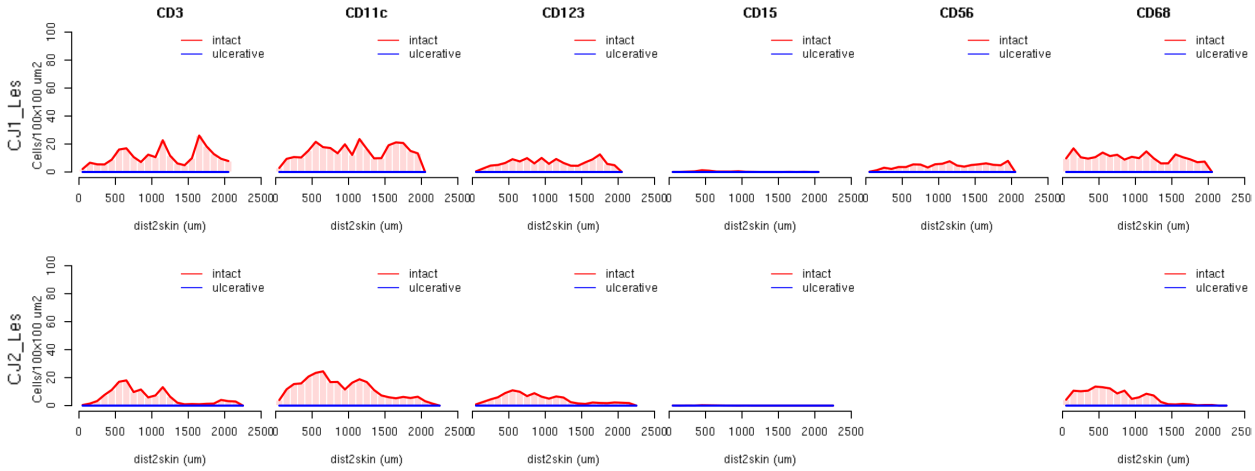
In this subsection, each figure shows all slides from a single biopsy.

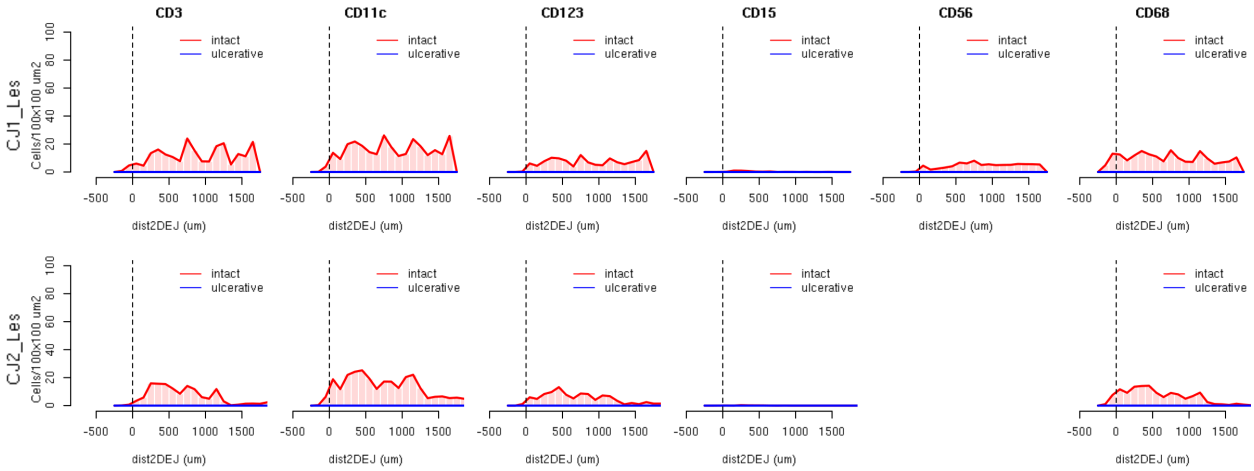

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
AS1_Les
AS2_Les
AS3_Les
Cells/100x100 um2
dist2skin (um)

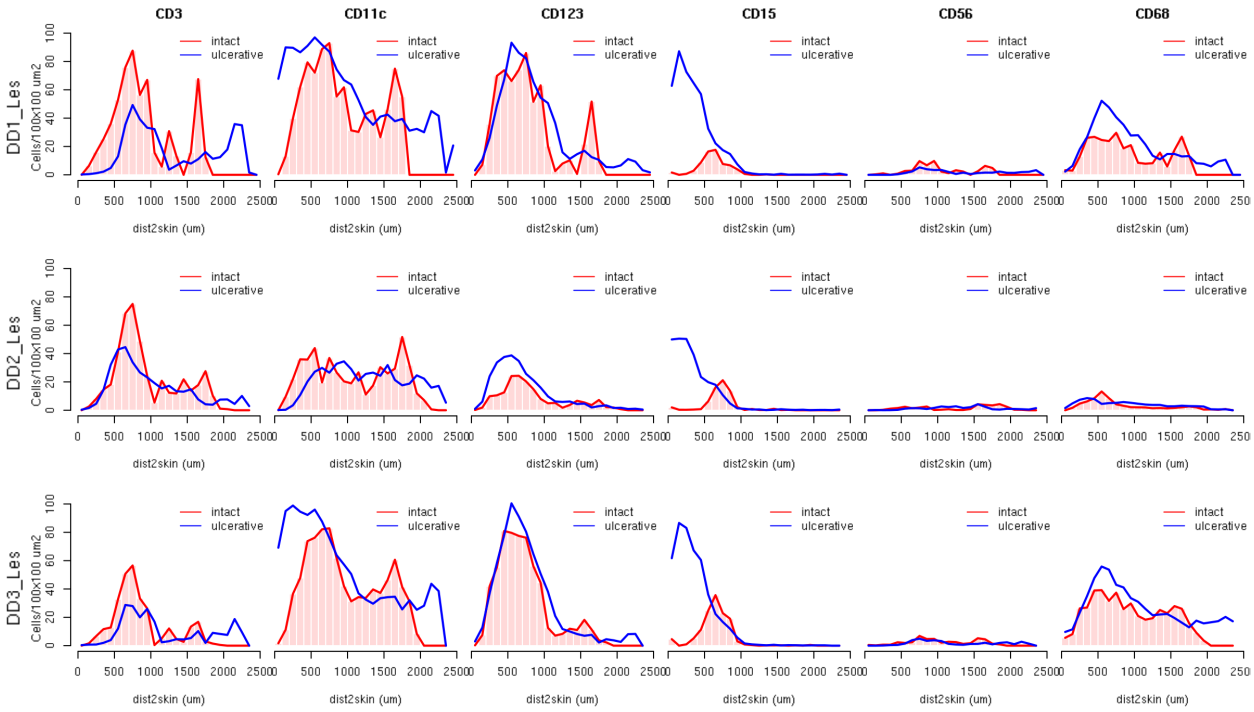

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
AS1_Les
AS2_Les
AS3_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

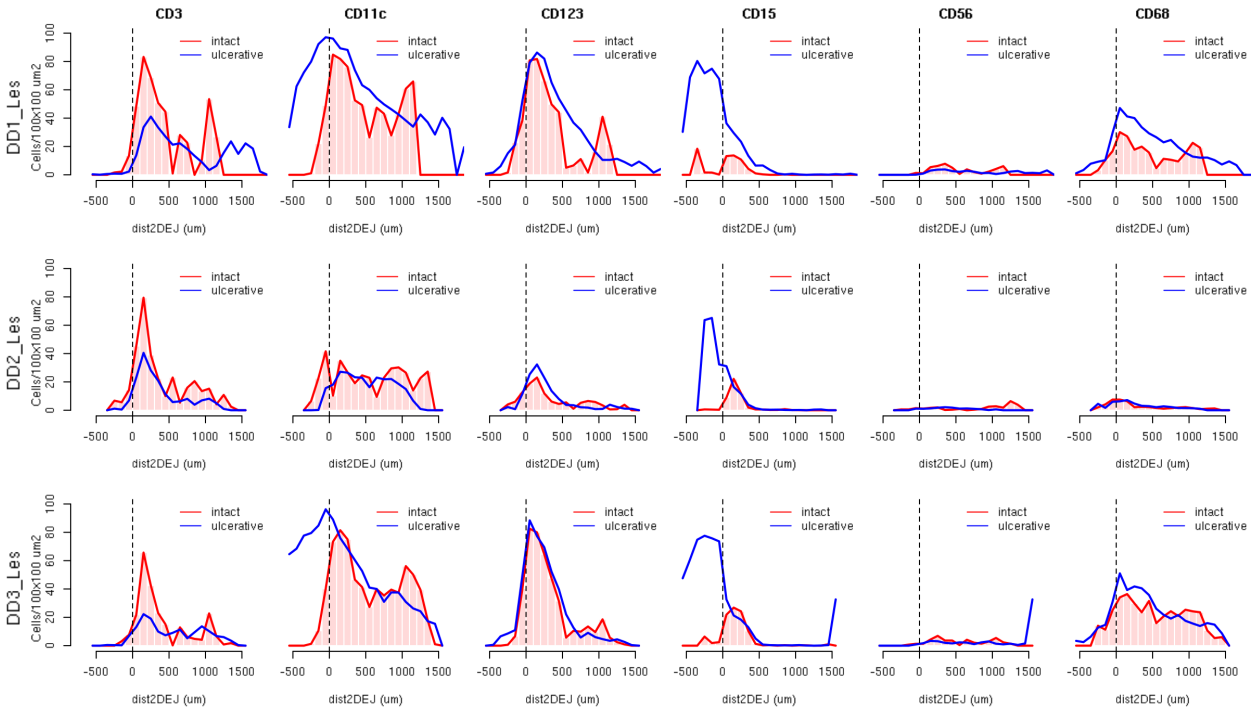

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
CF1_Les
CF2_Les
CF5_Les
Cells/100x100 um2
dist2skin (um)

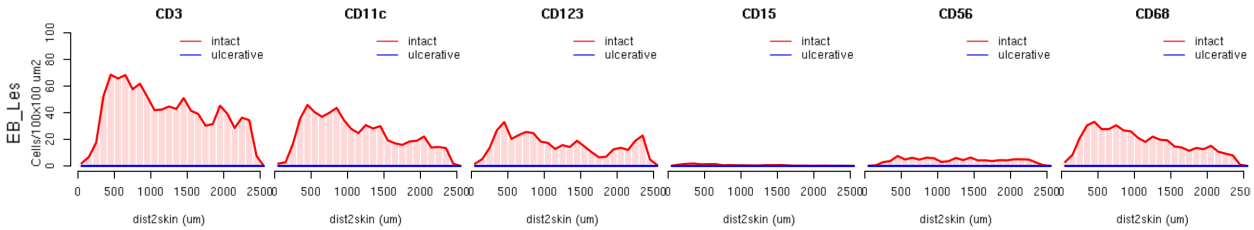

CD3
CD11c
CD123
CD15
CD56
CD68
CF1_Les
CF2_Les
CF5_Les
Cells/100x100 um2
intact
ulcerative
dist2DEJ (um)
-500
0
500
1000
1500
0
20
40
60
80
100

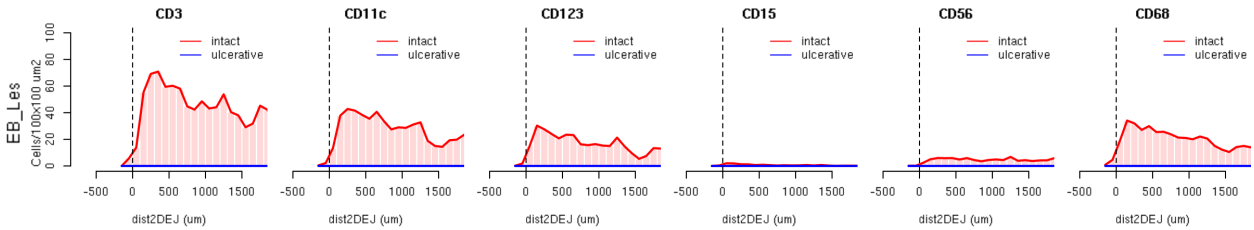

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
CJ1_Les
CJ2_Les
Cells/100x100 um2
dist2skin (um)

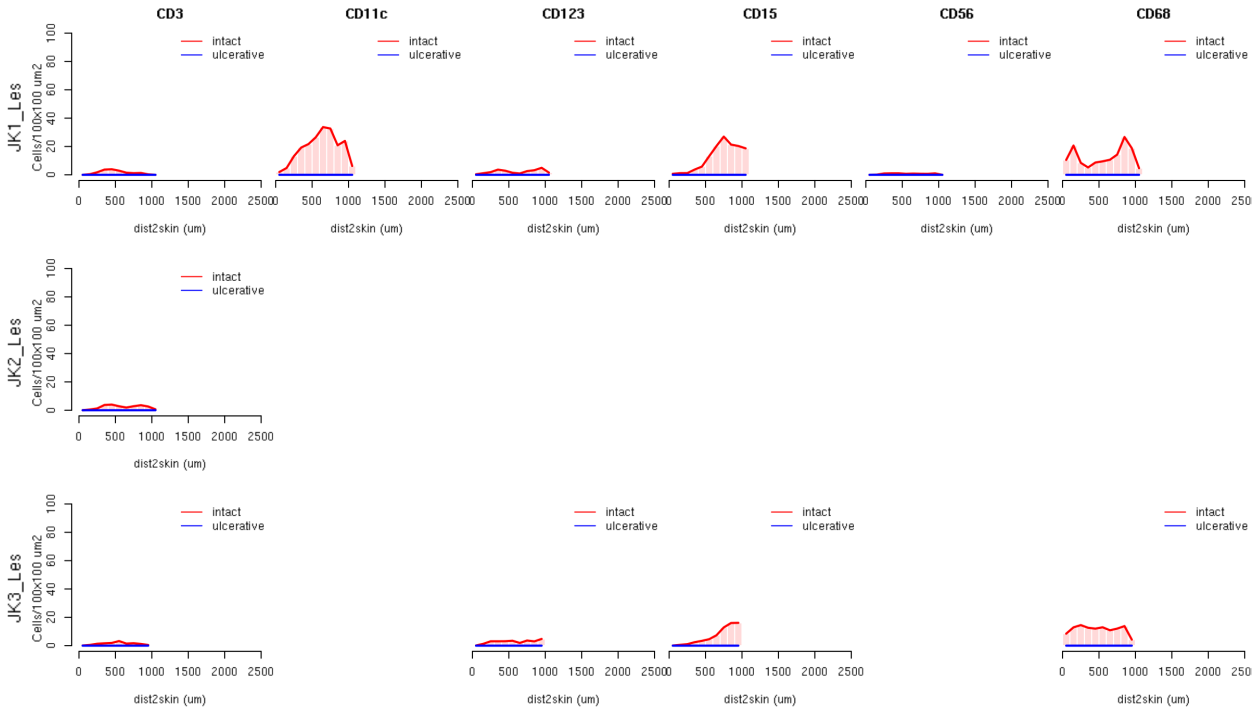

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
CJ1_Les
CJ2_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

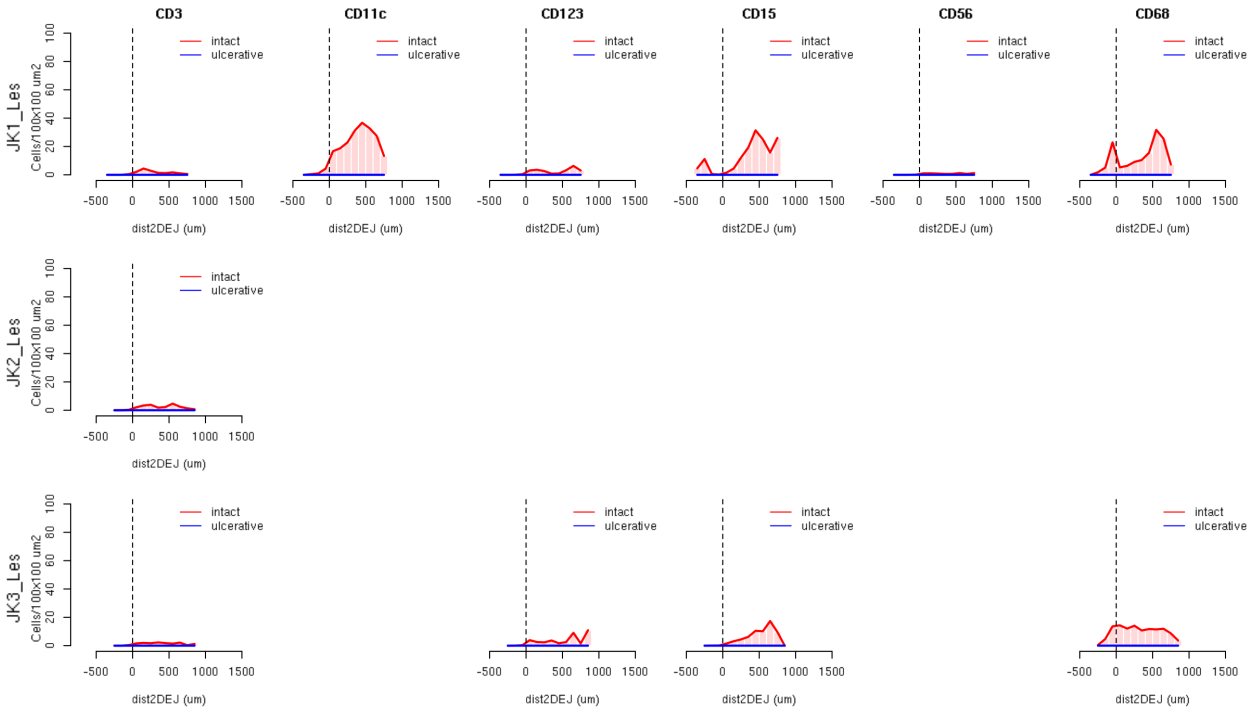

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
DD1_Les
DD2_Les
DD3_Les
Cells/100x100 um2
dist2skin (um)

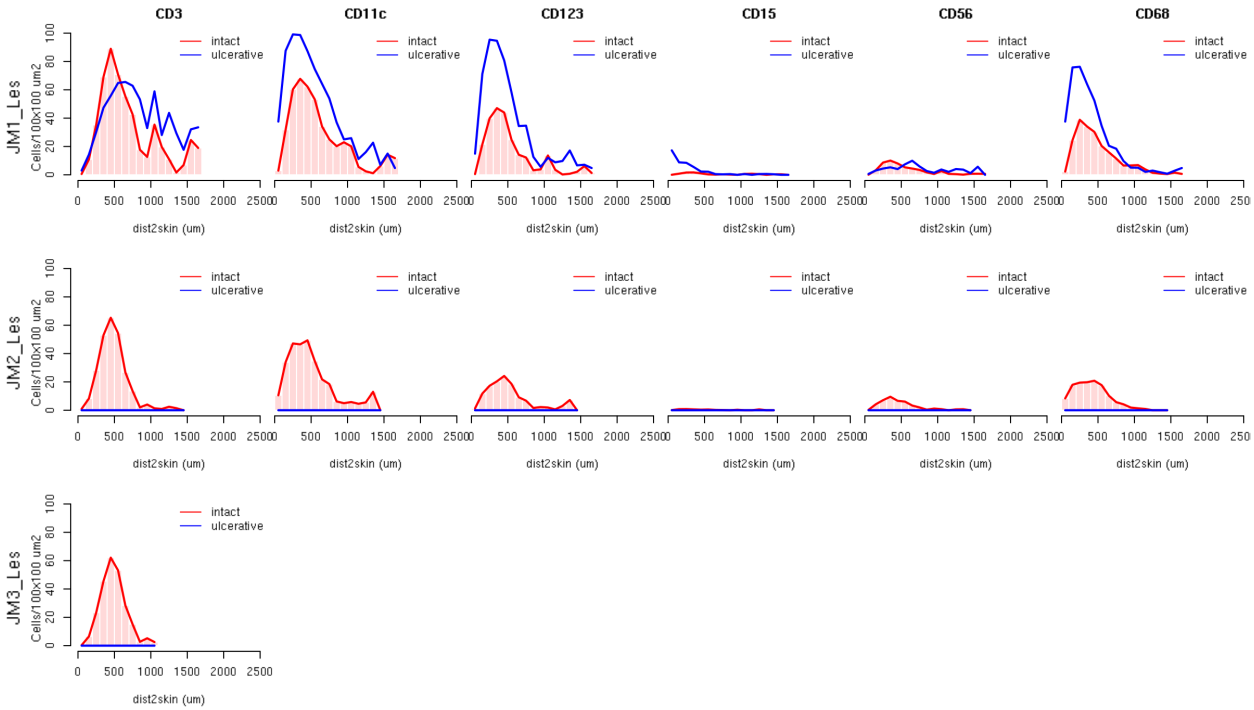

CD3
CD11c
CD123
CD15
CD56
CD68
DD1_Les
DD2_Les
DD3_Les
Cells/100x100 um2
intact
ulcerative
dist2DEJ (um)
-500
0
500
1000
1500
0
20
40
60
80
100

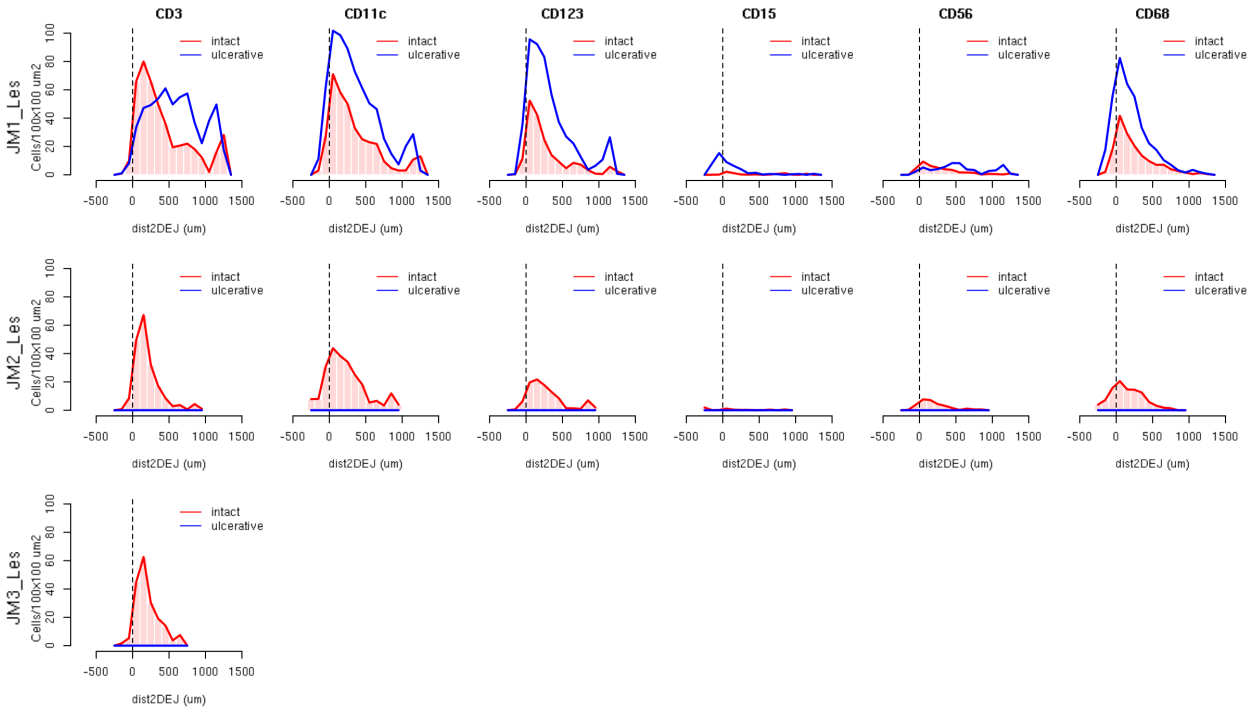

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
EB_Les
Cells/100x100 um2
0
20
40
60
80
100
0
500
1000
1500
2000
2500
dist2skin (um)

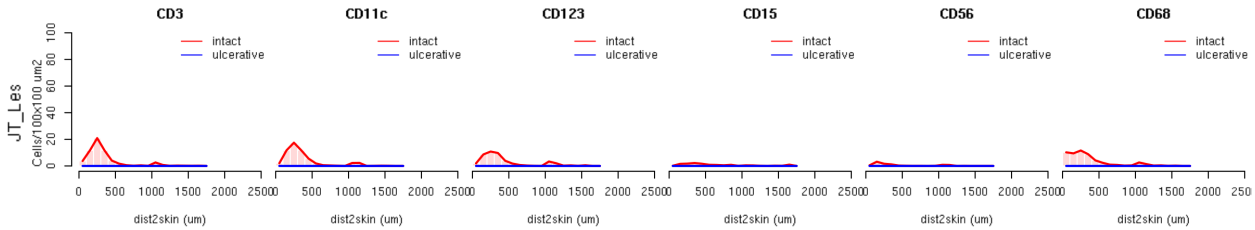

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
EB_Les
Cells/100x100 um2
0 20 40 60 80 100
-500 0 500 1000 1500
dist2DEJ (um)

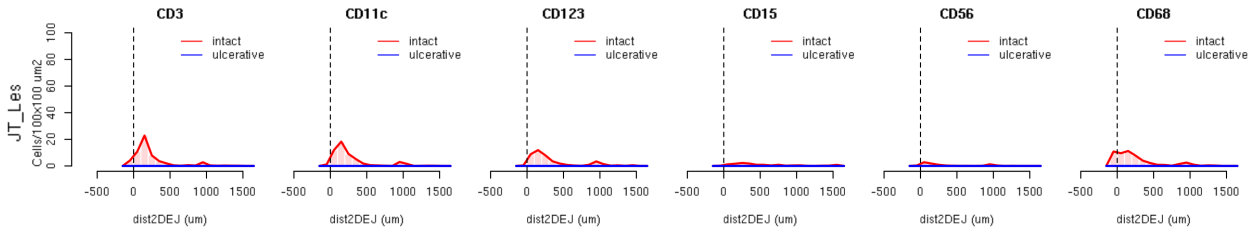

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JK1_Les
JK2_Les
JK3_Les
Cells/100x100 um2
dist2skin (um)

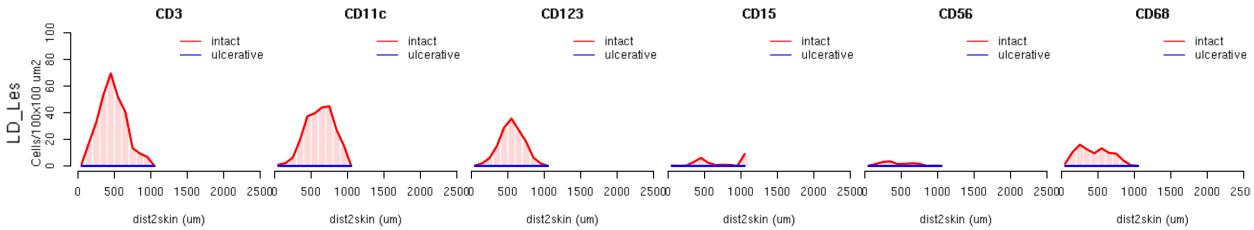

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JK1_Les
JK2_Les
JK3_Les
Cells/100x100 um2
dist2DEJ (um)

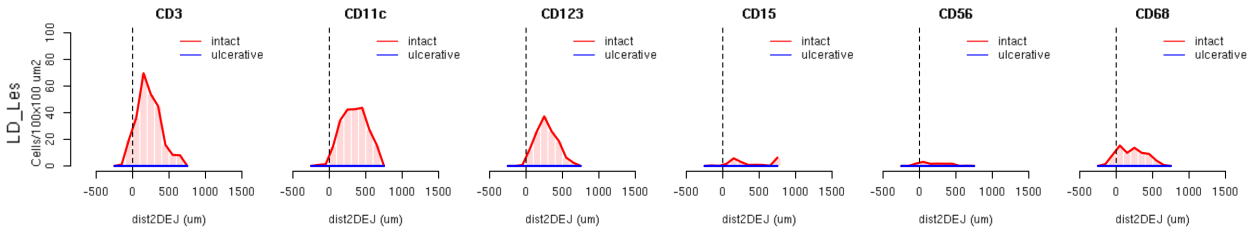

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JM1_Les
JM2_Les
JM3_Les
Cells/100x100 um2
dist2skin (um)

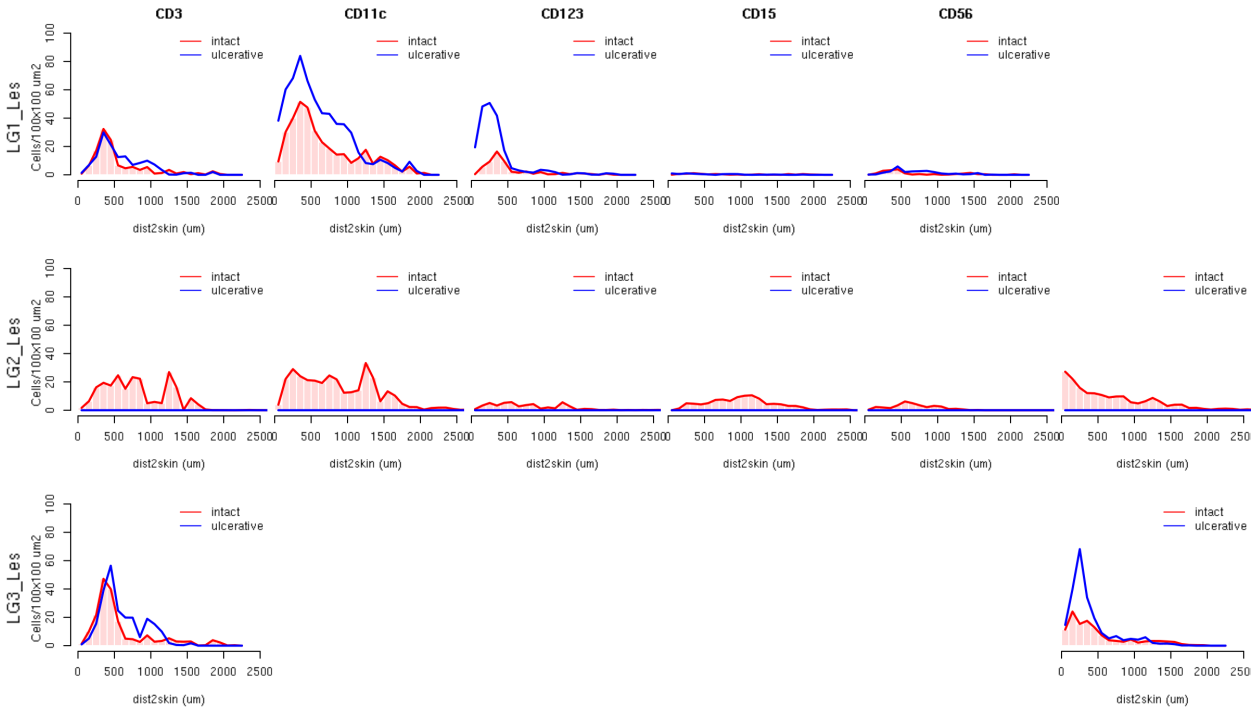

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JM1_Les
JM2_Les
JM3_Les
Cells/100x100 um2
dist2DEJ (um)

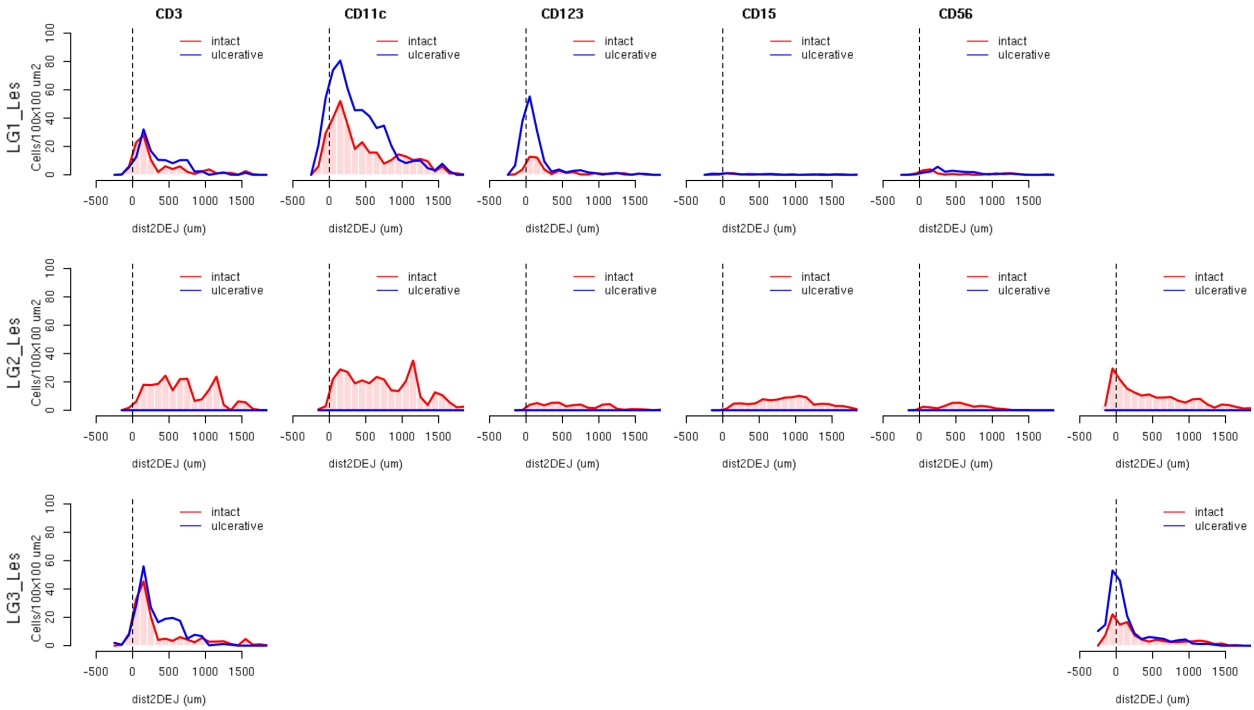

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JT_Les
Cells/100x100 um2
0
20
40
60
80
100
0
500
1000
1500
2000
2500
dist2skin (um)

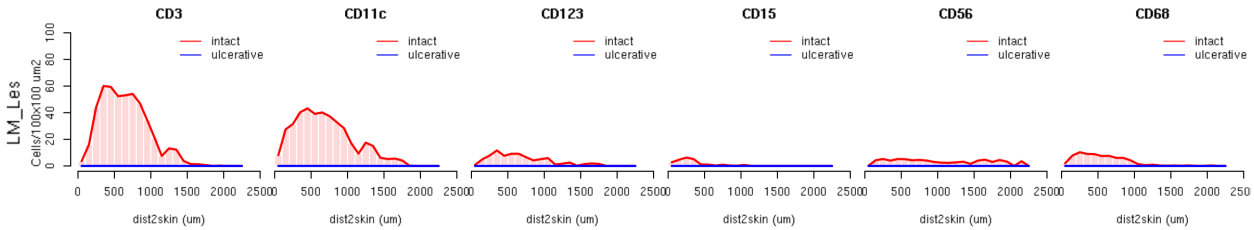

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JT_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

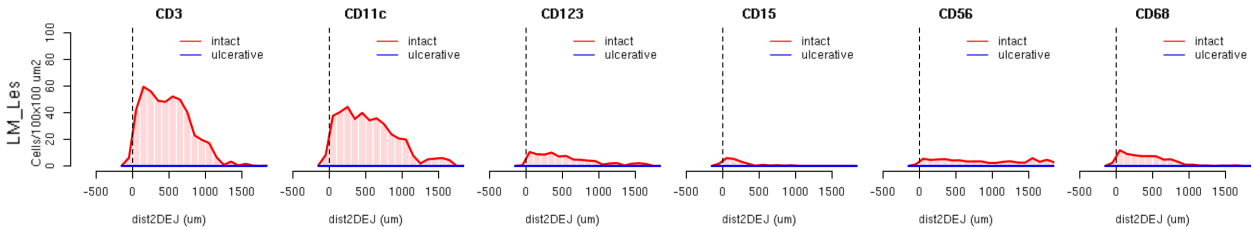

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LD_Les
Cells/100x100 um2
0
20
40
60
80
100
0
500
1000
1500
2000
2500
dist2skin (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LD_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

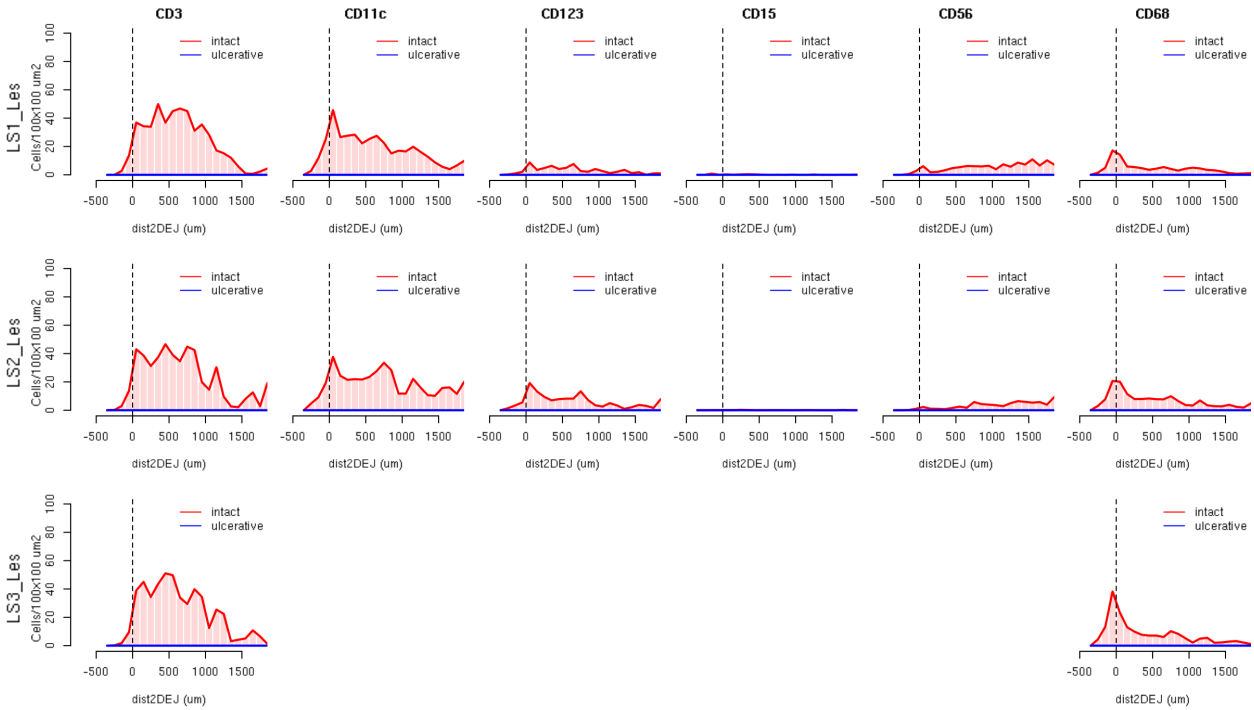

CD3
CD11c
CD123
CD15
CD56
intact
ulcerative
LG1_Les
LG2_Les
LG3_Les
Cells/100x100 um2
dist2skin (um)

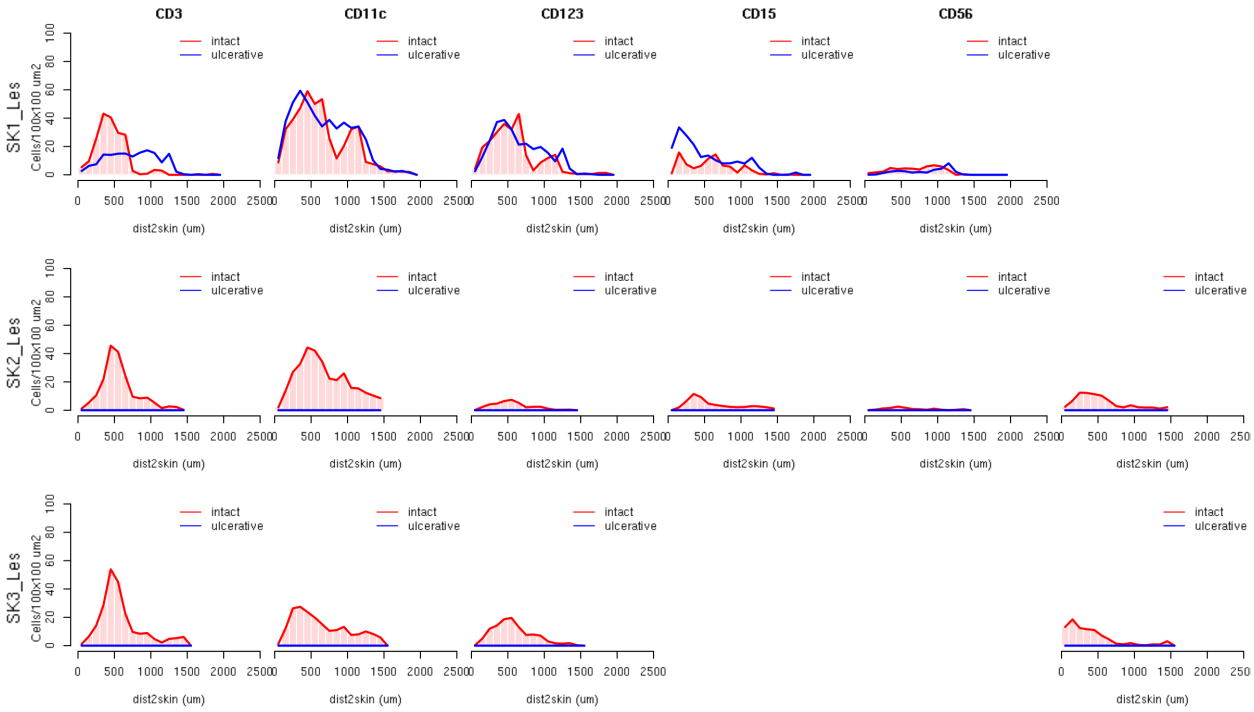

CD3
CD11c
CD123
CD15
CD56
intact
ulcerative
LG1_Les
LG2_Les
LG3_Les
Cells/100x100 um2
dist2DEJ (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LM_Les
Cells/100x100 um2
0
20
40
60
80
100
0
500
1000
1500
2000
2500
dist2skin (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LM_Les
Cells/100x100 um2
0 20 40 60 80 100
-500 0 500 1000 1500
dist2DEJ (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LS1_Les
LS2_Les
LS3_Les
Cells/100x100 um2
dist2skin (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LS1_Les
LS2_Les
LS3_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

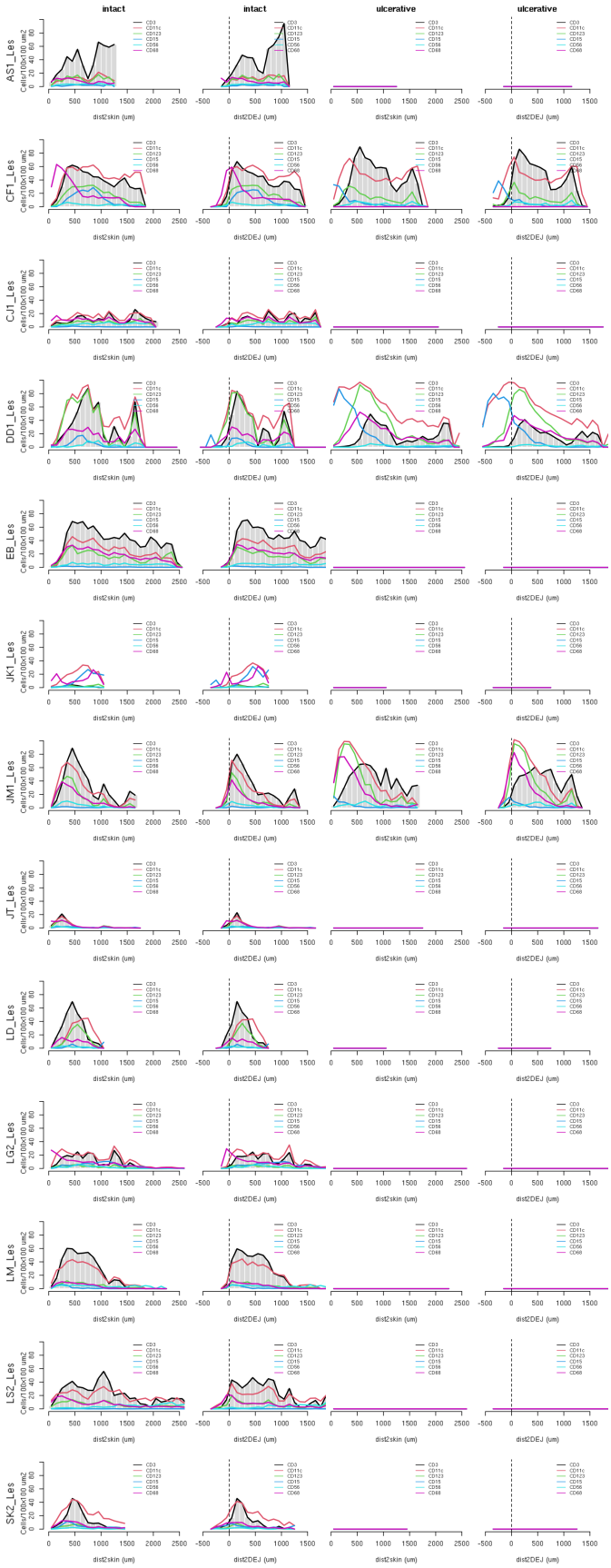

CD3
CD11c
CD123
CD15
CD56
intact
ulcerative
SK1_Les
SK2_Les
SK3_Les
Cells/100x100 um2
dist2skin (um)


CD3
CD11c
CD123
CD15
CD56
SK1_Les
SK2_Les
SK3_Les
Cells/100x100 um2
0 20 40 60 80 100
intact
ulcerative
-500 0 500 1000 1500
dist2DEJ (um)

Markers comparison - select slides

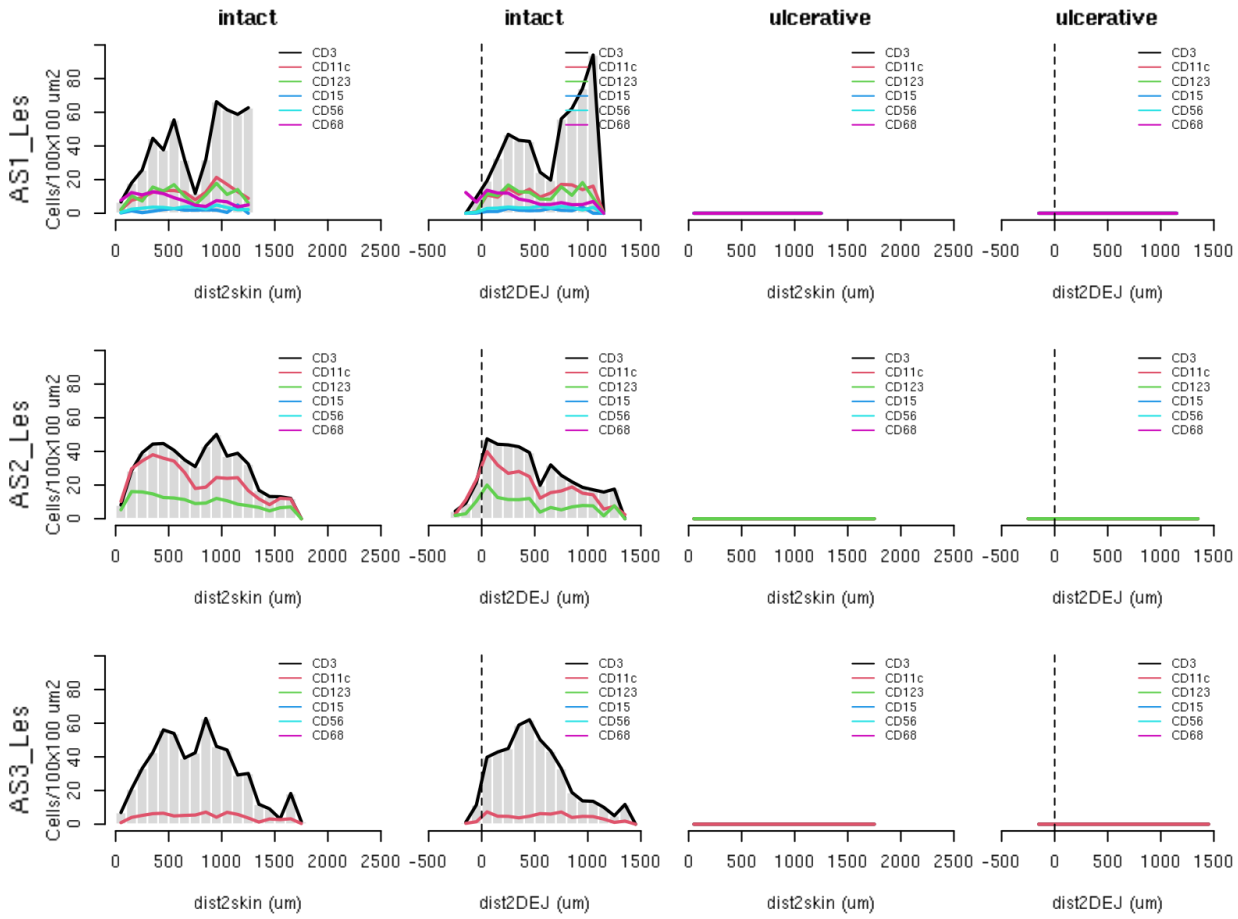
In this subsection, there is one figure showing 13 representative slides. Some observations:

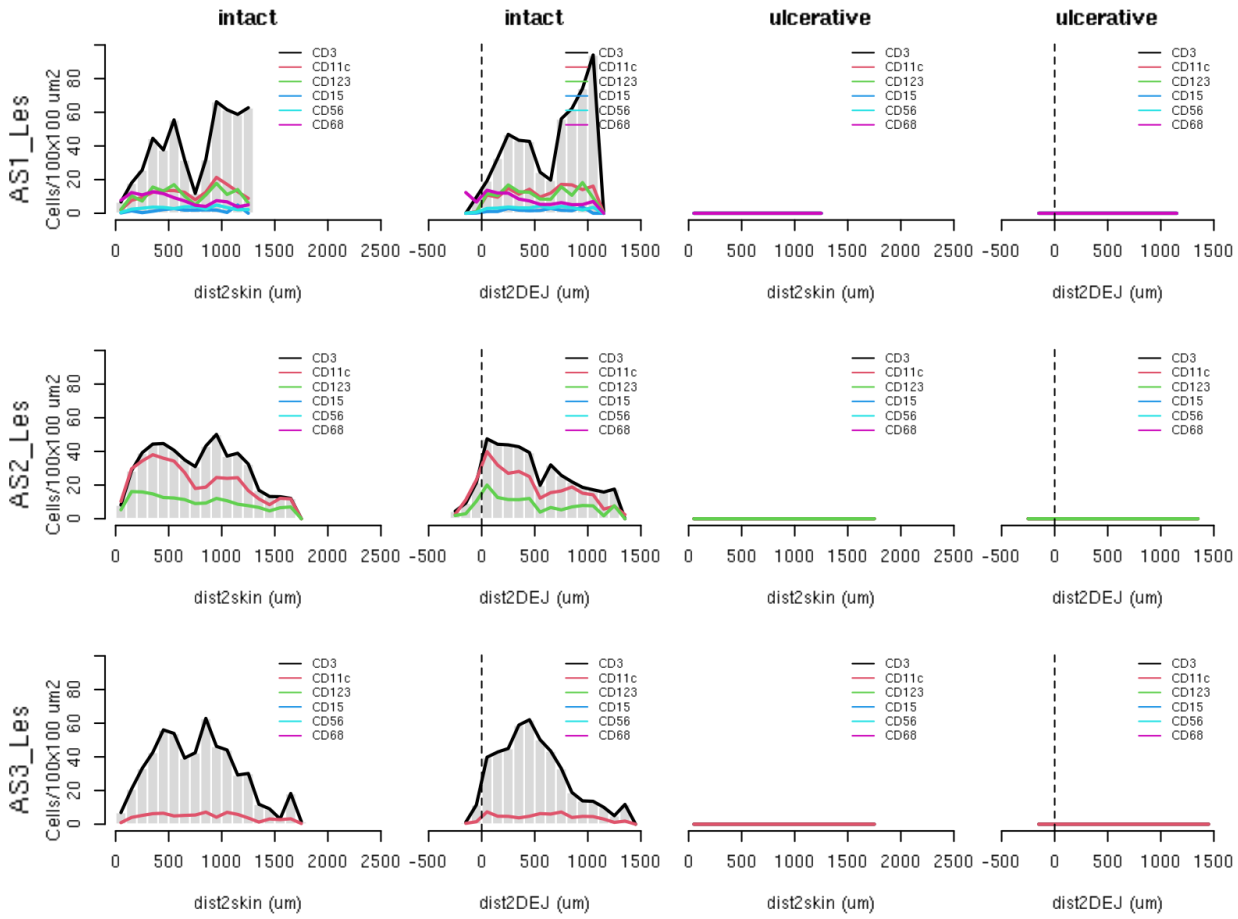
- The CD15 dist2skin peak in DD/lesion is near skin while the corresponding dist2DEJ peak is near DEJ is due to the fact that the DEJ line is very curvy.

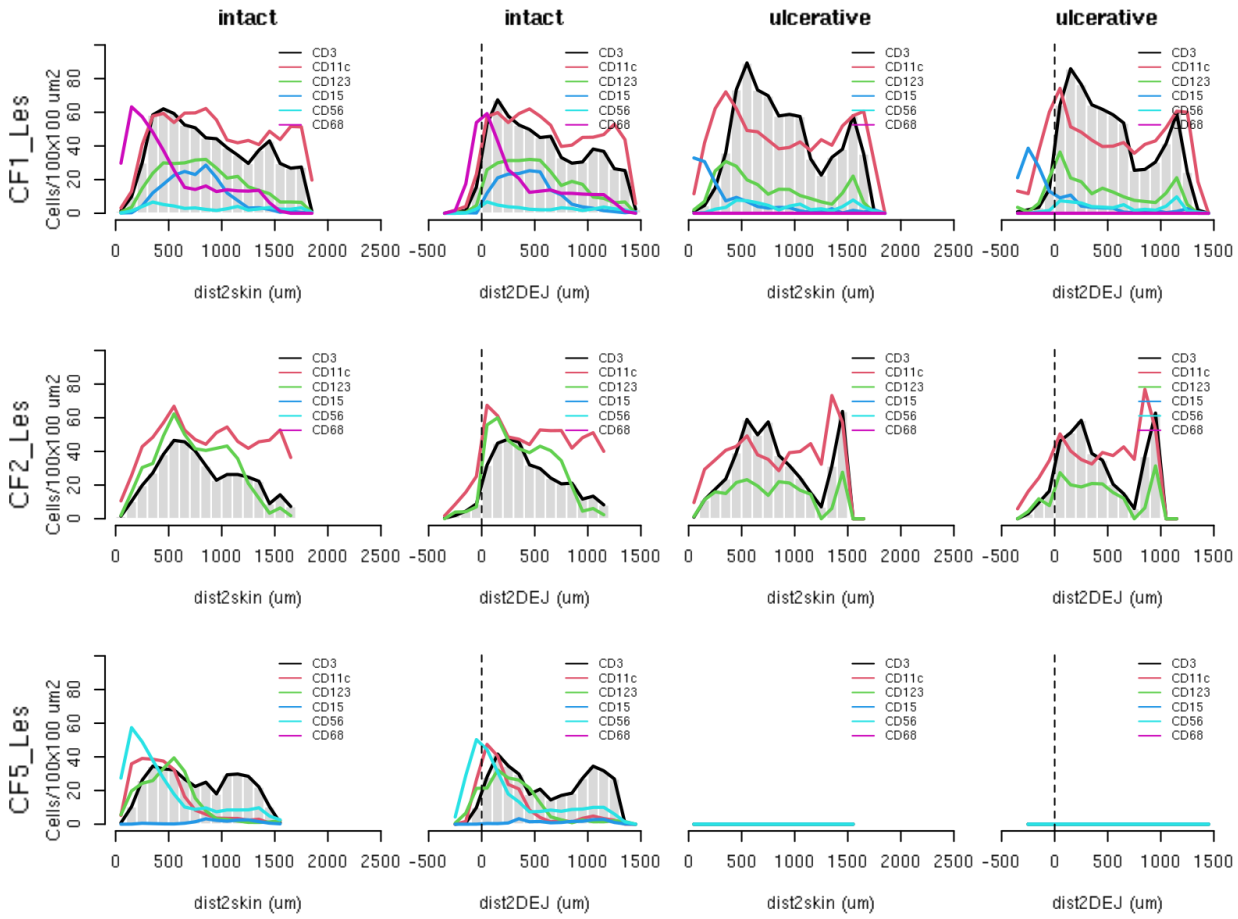

intact
intact
ulcerative
ulcerative
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
Cells/100um^2
dist2skin (um)
dist2DEJ (um)
CD3
CD11c
CD123
CD15
CD56
CD68

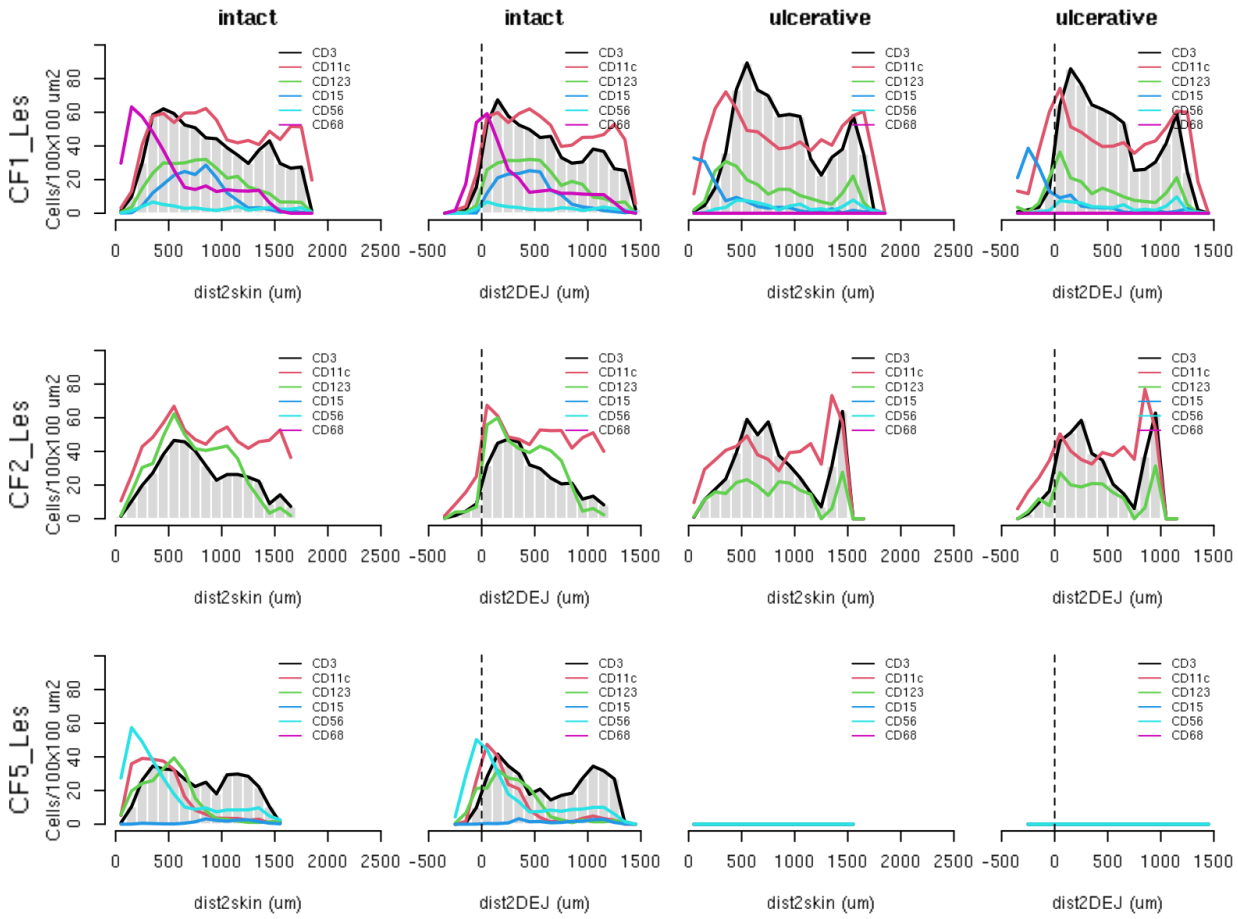
Markers comparison - all slides

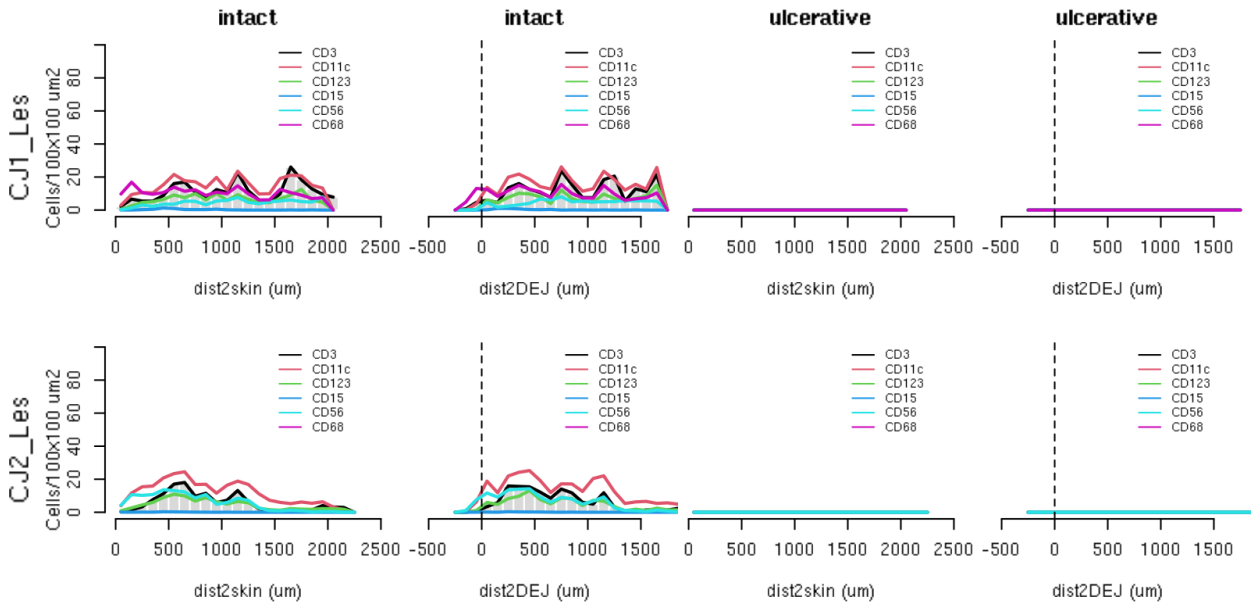
In this subsection, each figure shows all slides from a single biopsy.

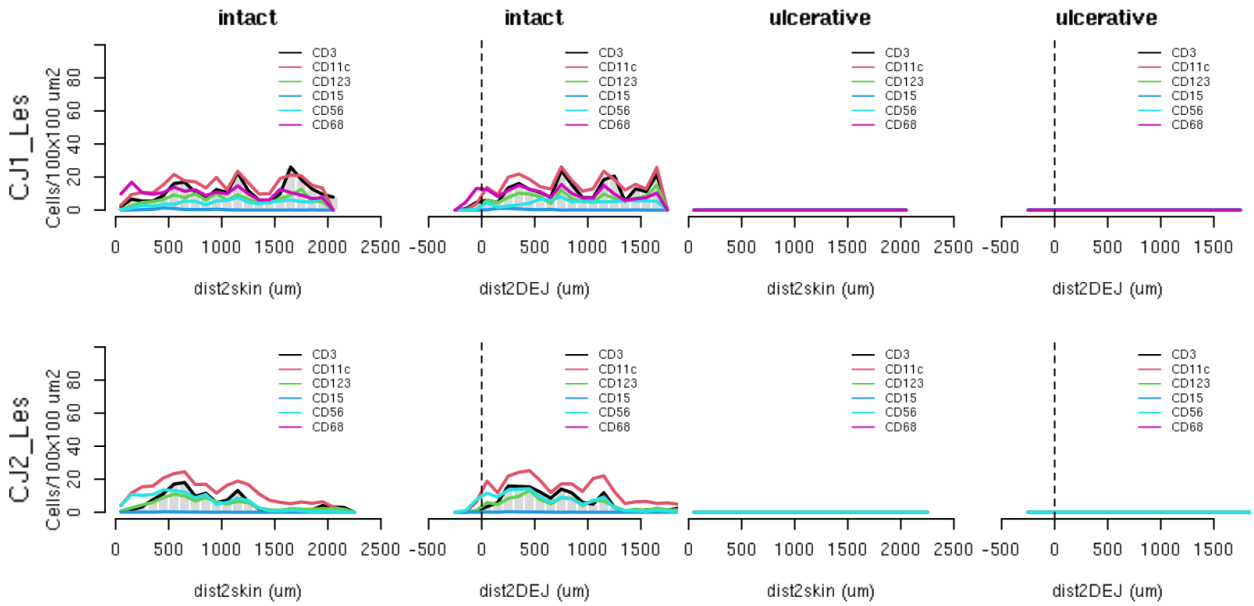

intact
intact
ulcerative
ulcerative
AS1_Les
AS2_Les
AS3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

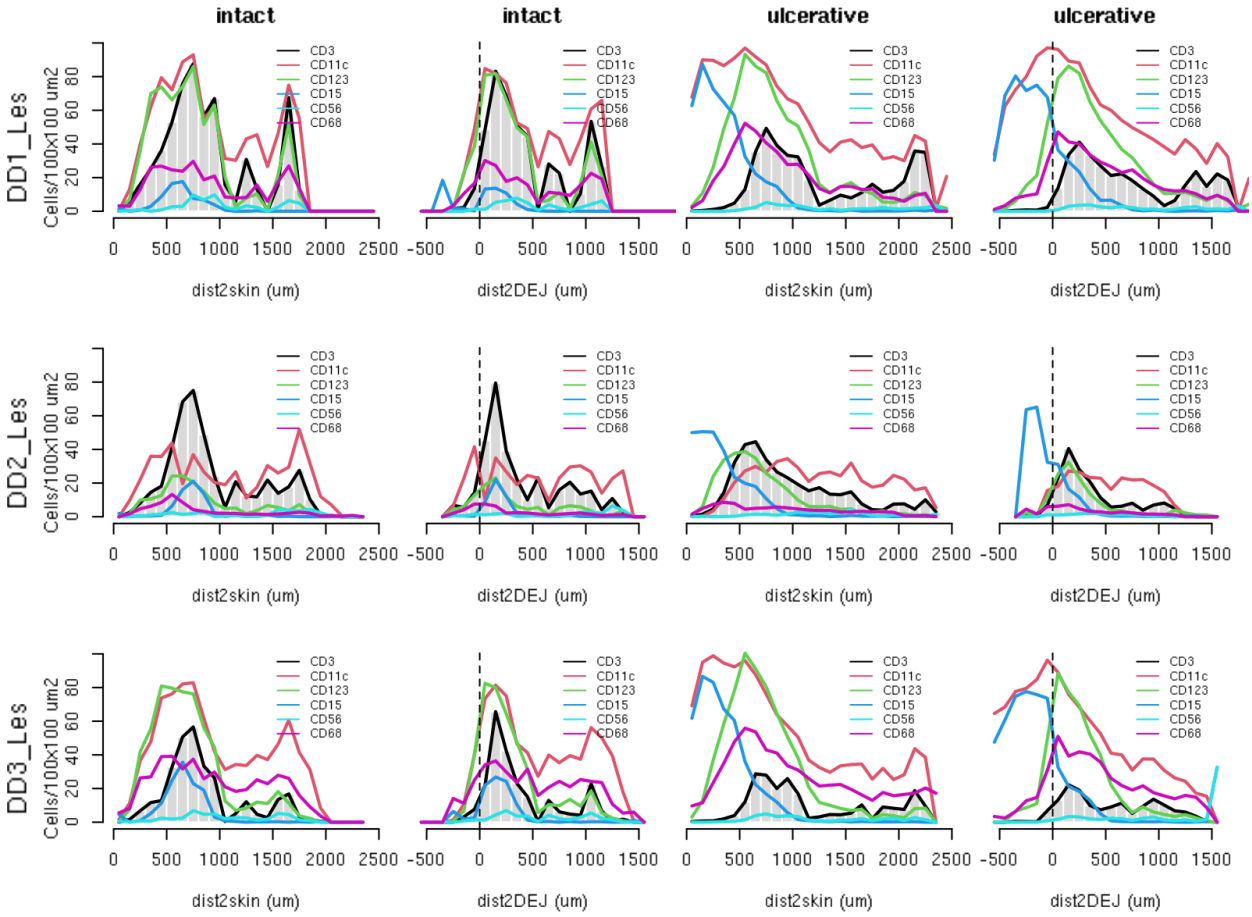

intact
intact
ulcerative
ulcerative
AS1_Les
AS2_Les
AS3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

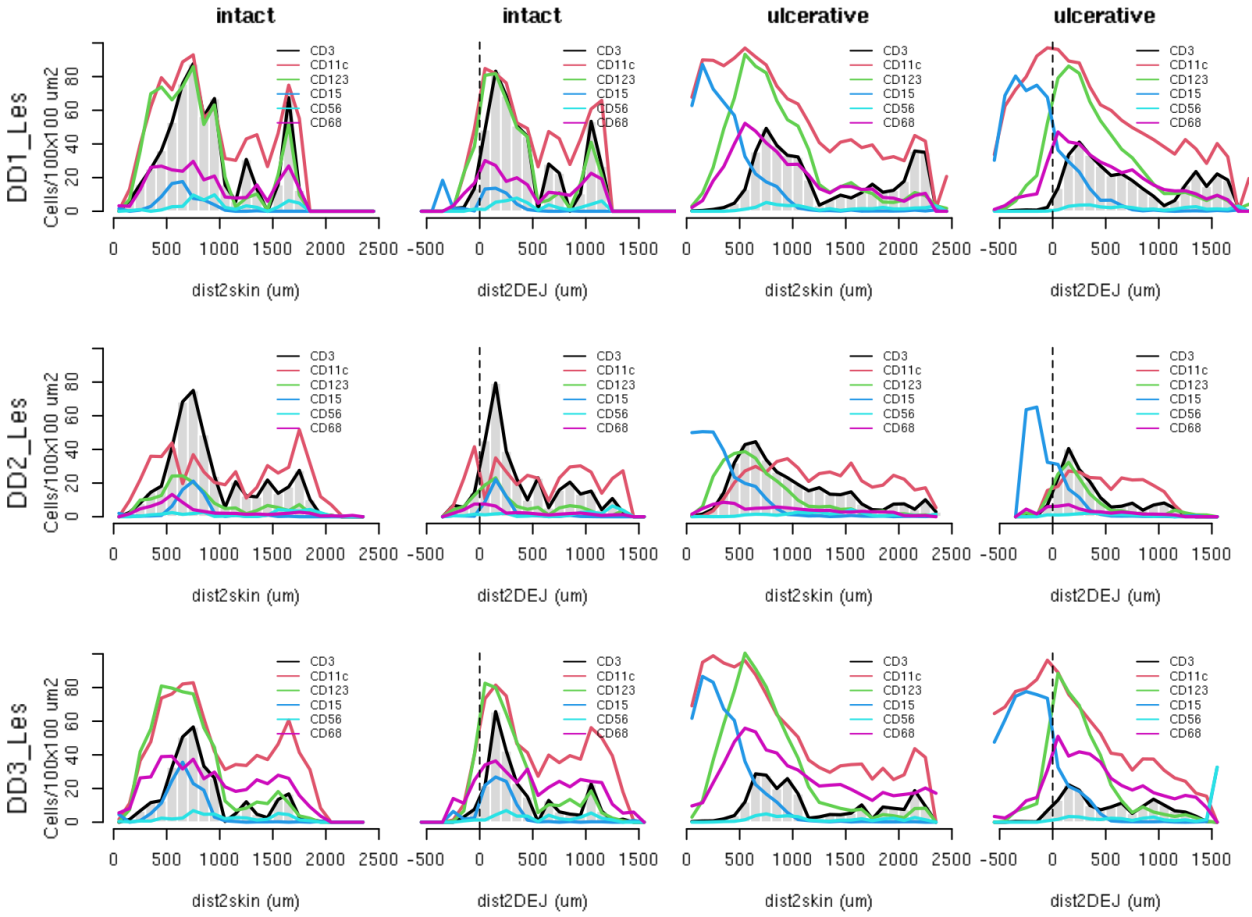

intact
intact
ulcerative
ulcerative
CF1_Les
CF2_Les
CF5_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

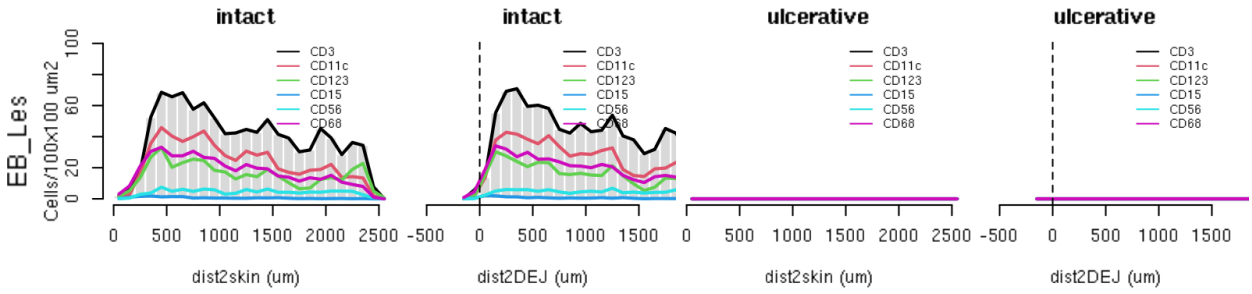

intact
intact
ulcerative
ulcerative
CF1_Les
CF2_Les
CF5_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

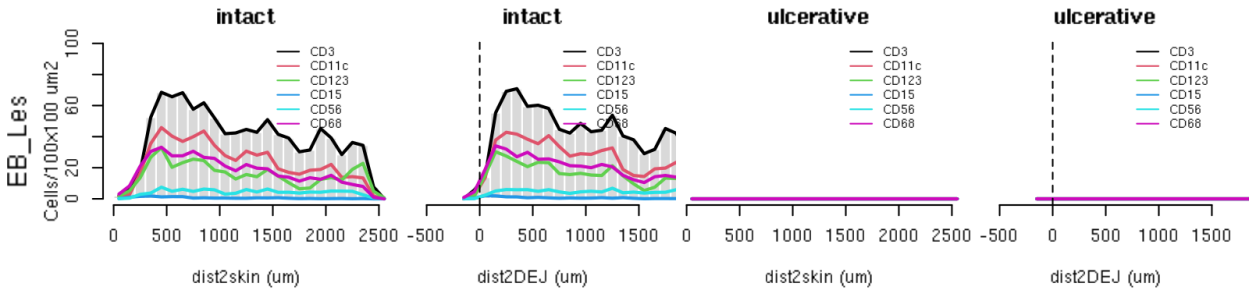

intact
intact
ulcerative
ulcerative
CJ1_Les
CJ2_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

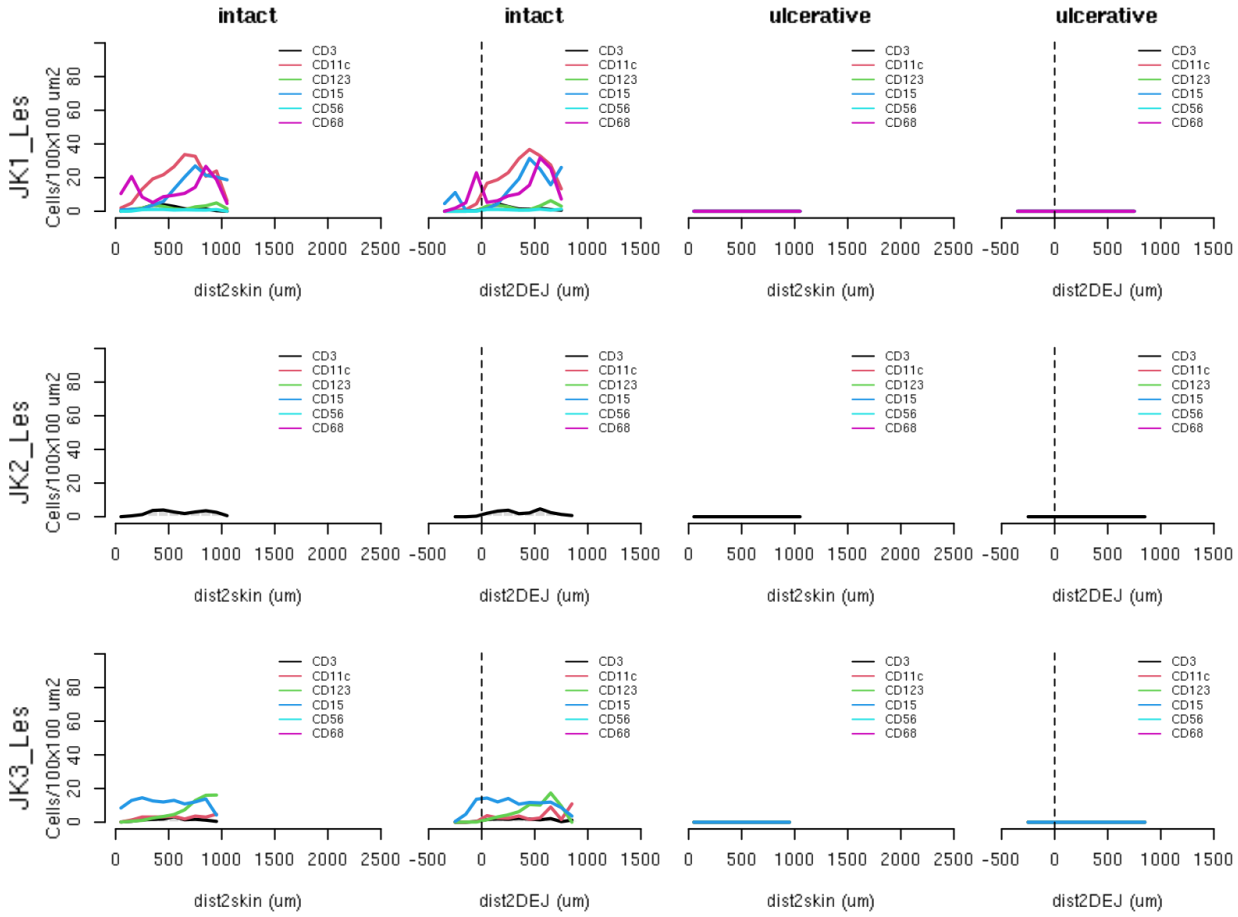

intact
intact
ulcerative
ulcerative
CJ1_Les
CJ2_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

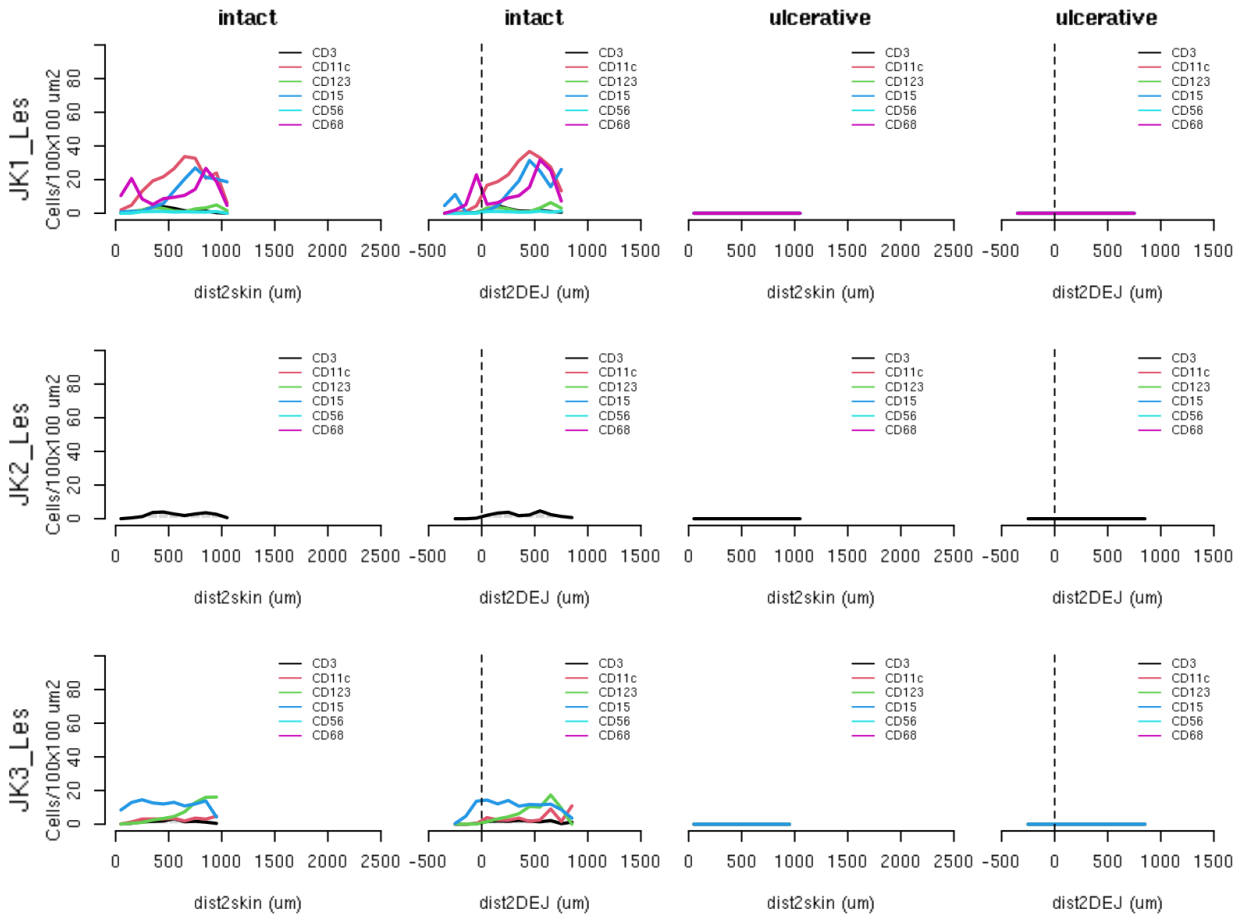

intact
intact
ulcerative
ulcerative
DD1_Les
DD2_Les
DD3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

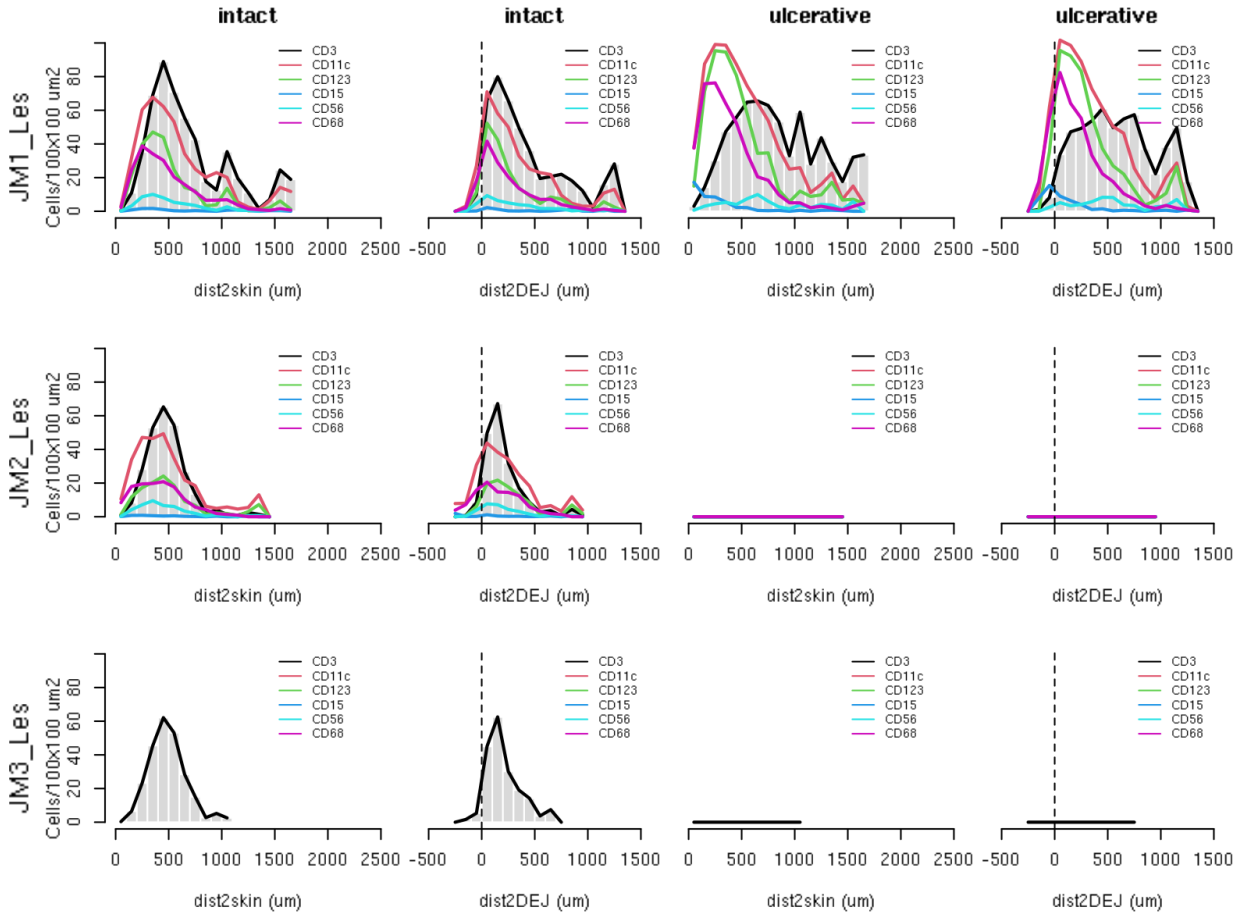

intact
intact
ulcerative
ulcerative
DD1_Les
DD2_Les
DD3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

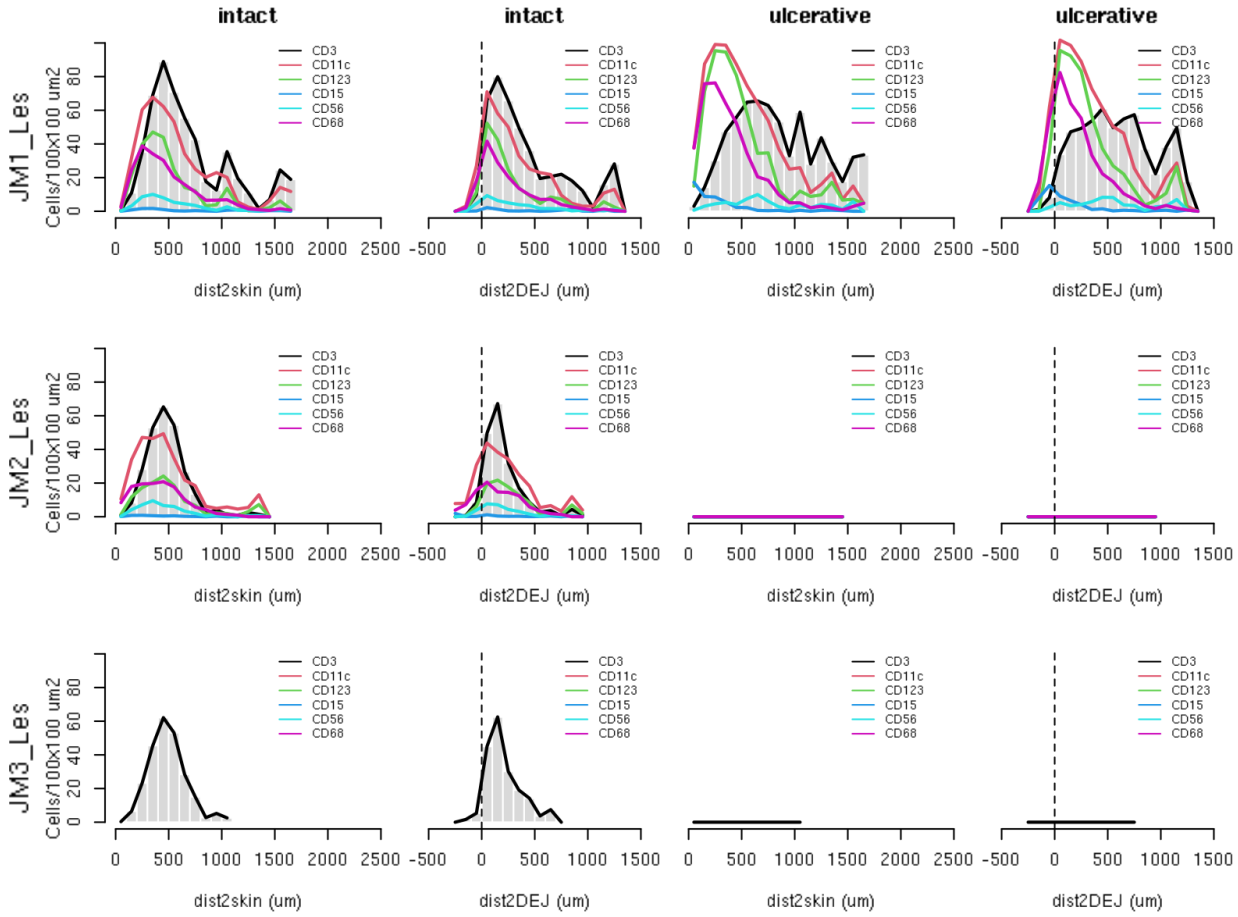

intact
intact
ulcerative
ulcerative
EB_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)

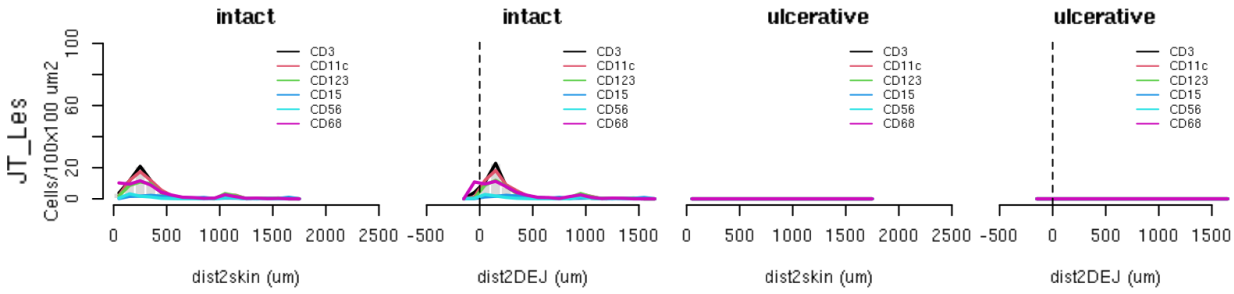

intact
intact
ulcerative
ulcerative
EB_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)

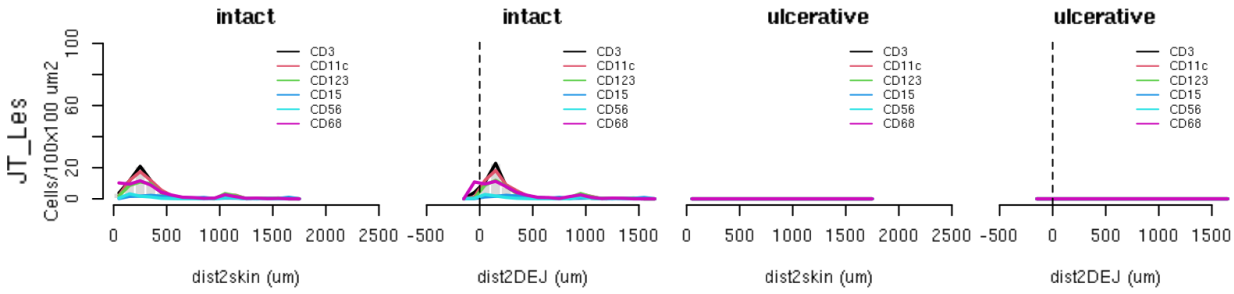

intact
intact
ulcerative
ulcerative
JK1_Les
JK2_Les
JK3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

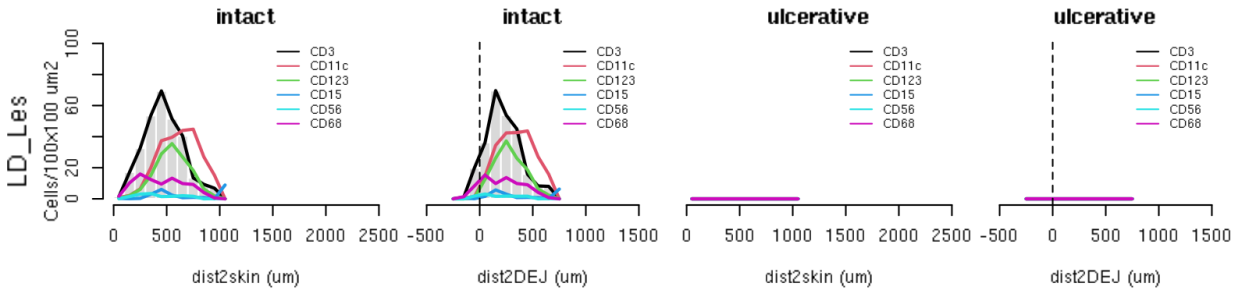

intact
intact
ulcerative
ulcerative
JK1_Les
JK2_Les
JK3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

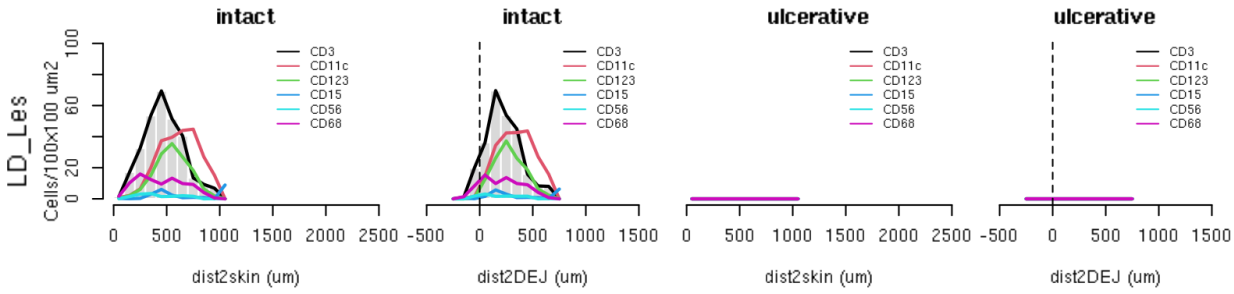

intact
intact
ulcerative
ulcerative
JM1_Les
JM2_Les
JM3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

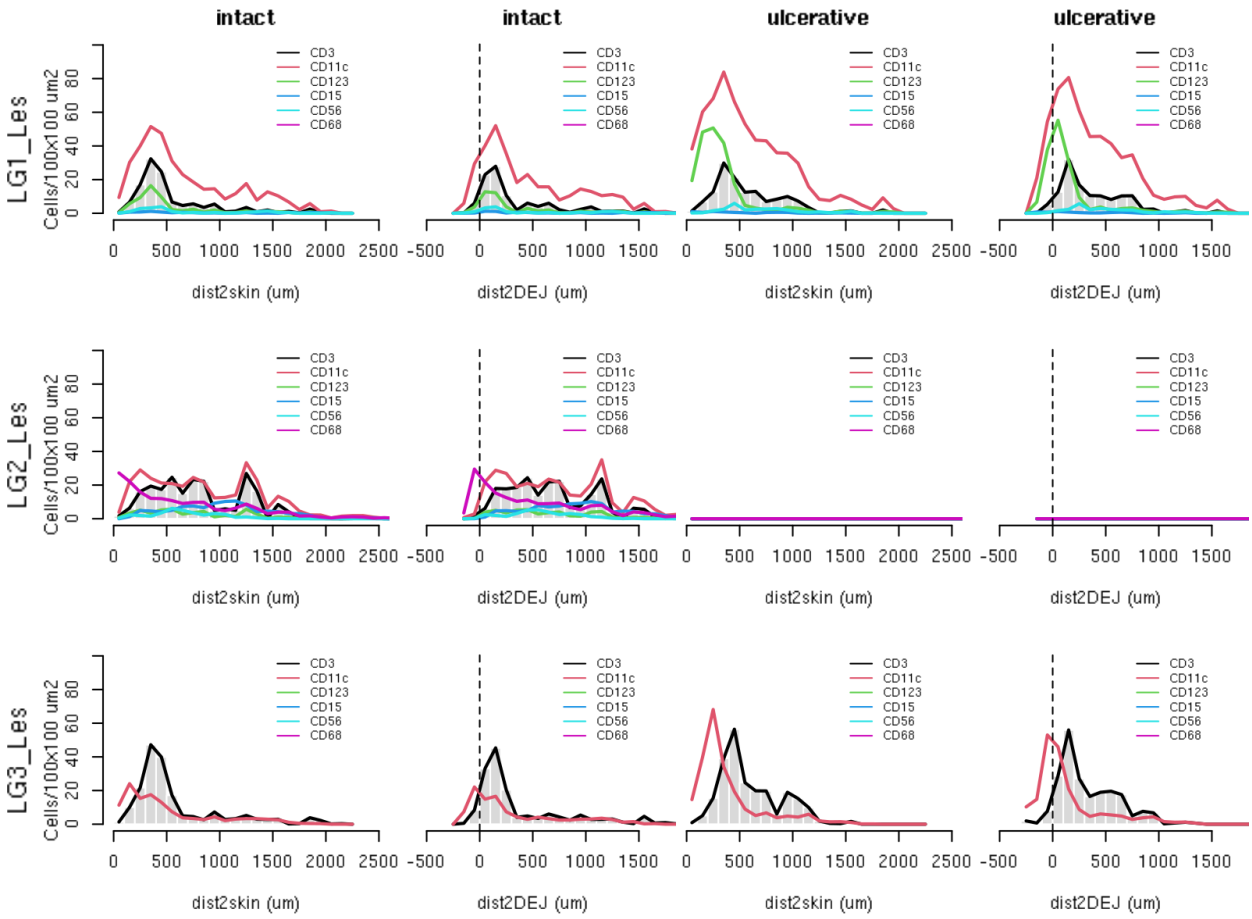

intact
intact
ulcerative
ulcerative
JM1_Les
JM2_Les
JM3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

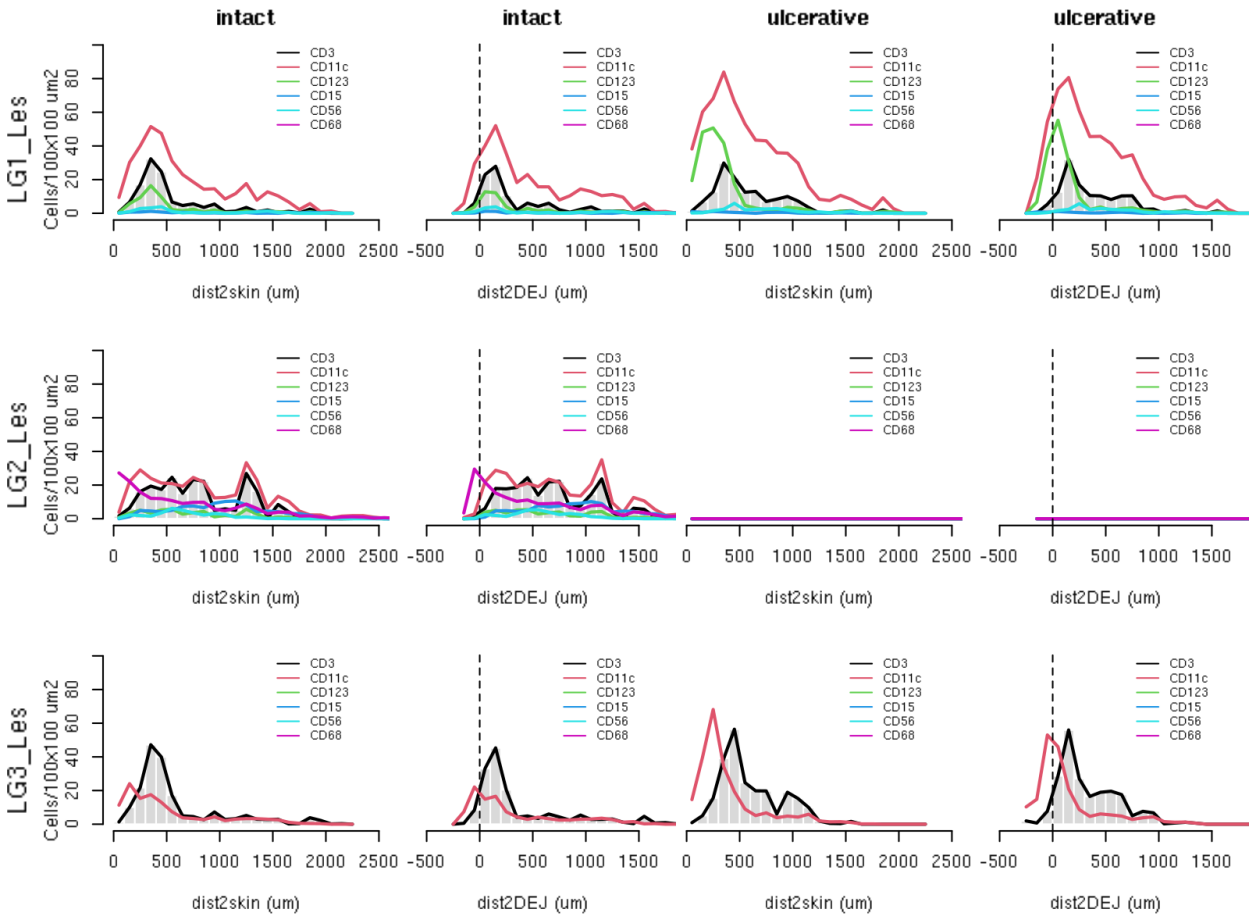

intact
intact
ulcerative
ulcerative
JT_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)

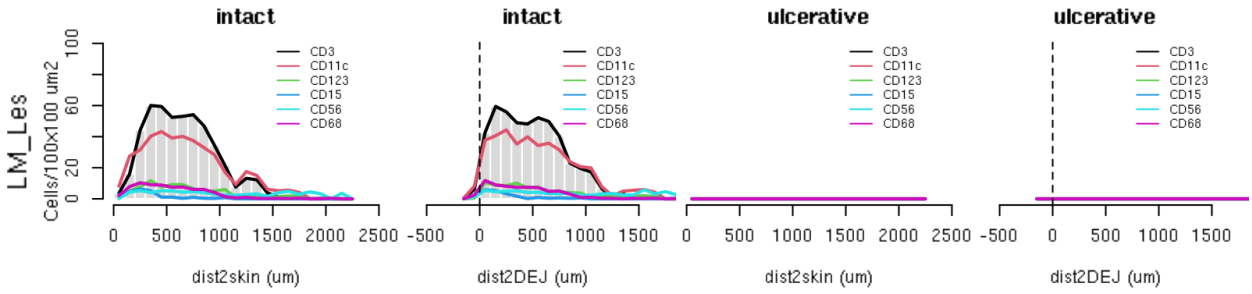

intact
intact
ulcerative
ulcerative
JT_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)

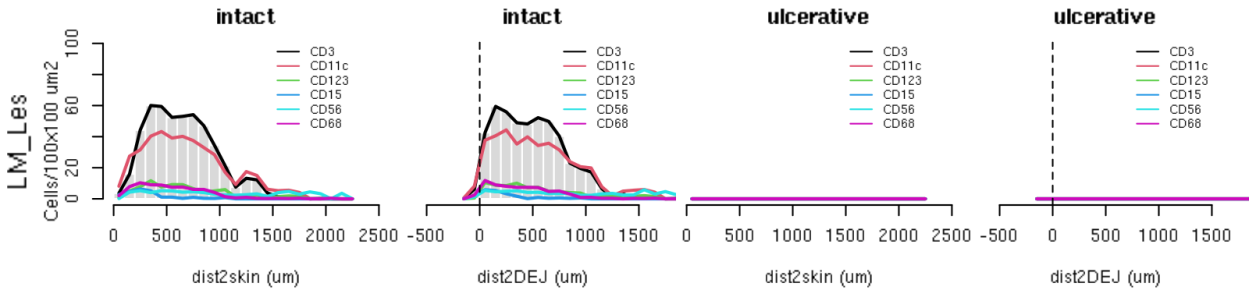

intact
intact
ulcerative
ulcerative
LD_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)

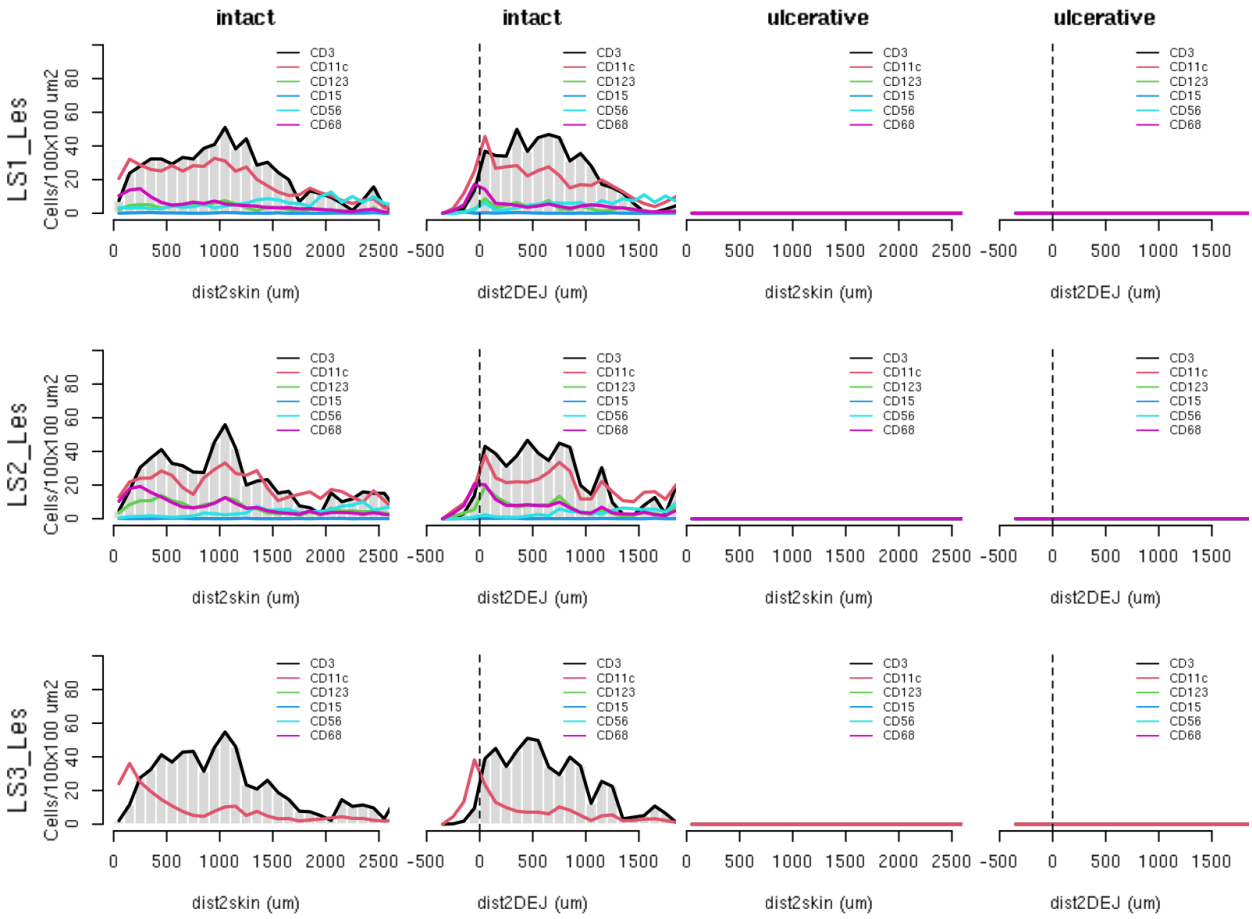

intact
intact
ulcerative
ulcerative
LD_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)

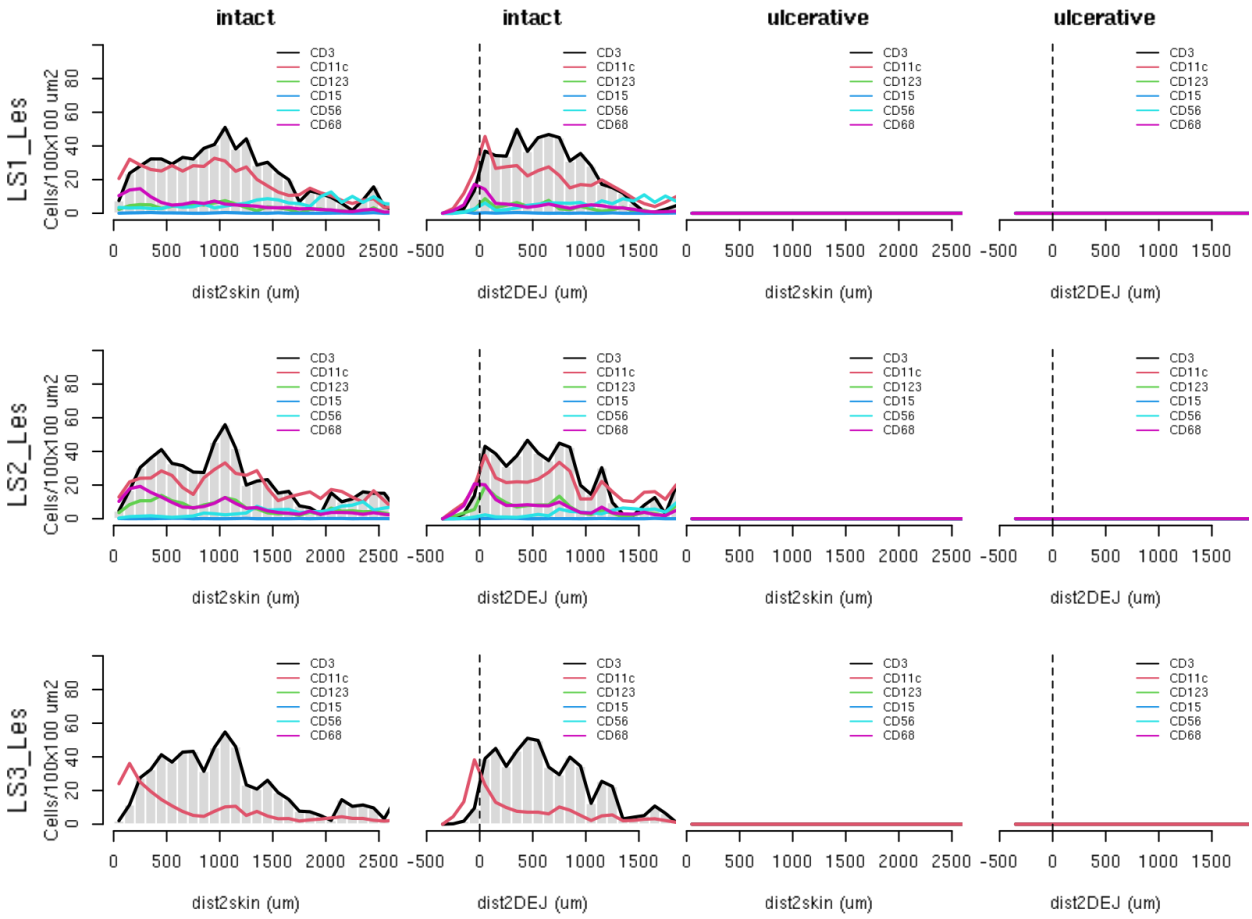

intact
intact
ulcerative
ulcerative
LG1_Les
LG2_Les
LG3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

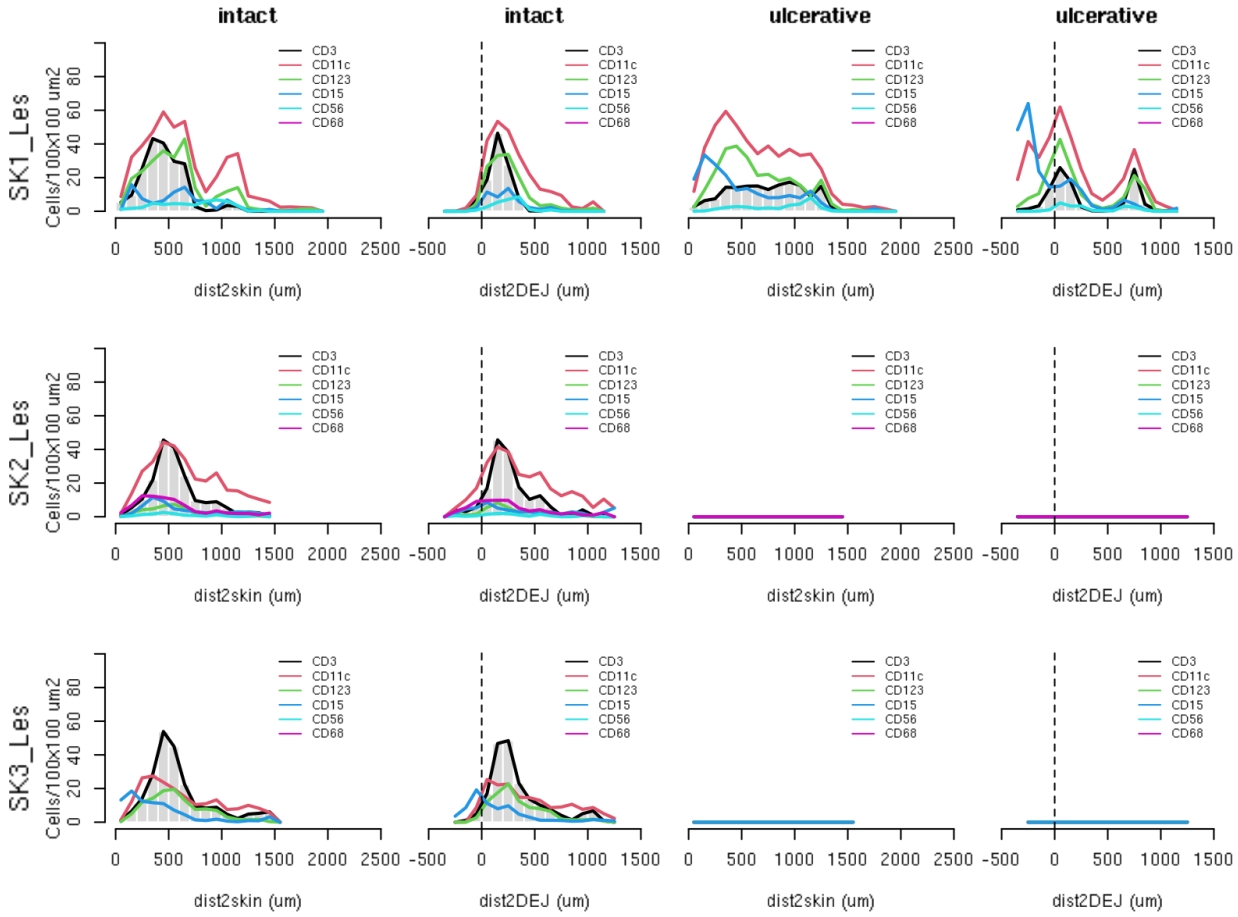

intact
intact
ulcerative
ulcerative
LG1_Les
LG2_Les
LG3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

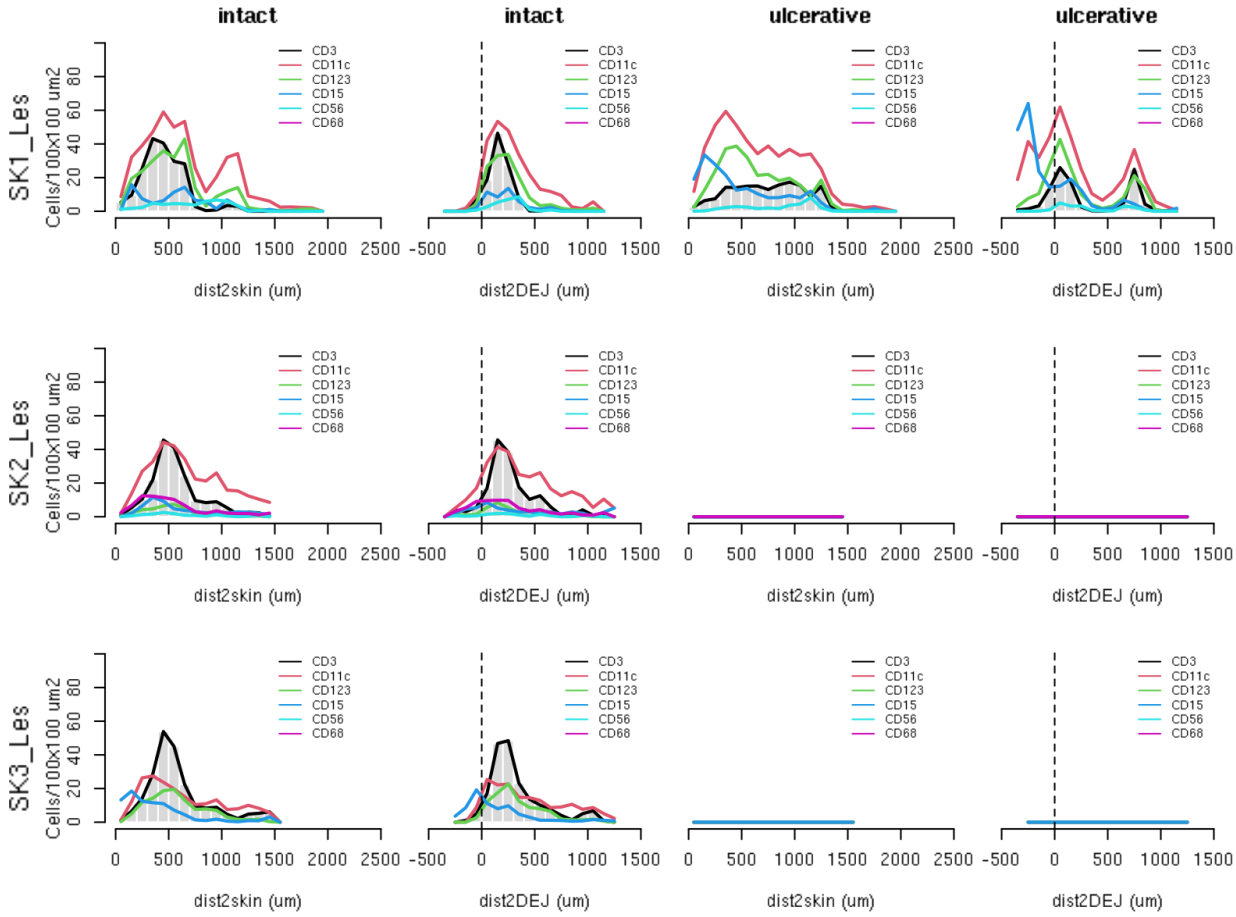

intact
intact
ulcerative
ulcerative
LM_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
LM_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
LS1_Les
LS2_Les
LS3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
LS1_Les
LS2_Les
LS3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
SK1_Les
SK2_Les
SK3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
SK1_Les
SK2_Les
SK3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

Count tables

For each slide, there are two tables, one for up to 500 um below DEJ and one for up to 1000 um below DEJ.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	18	1372	0	0	90	1575			92	11968	0	0
CD123	15	1353	0	0	75	1553			76	11401	0	0
CD15	2	195	0	0	10	223			10	1887	0	0
CD279	69	1453	0	0	348	1668			354	15023	0	0
CD3	187	4071	0	0	944	4673			959	40233	0	0
CD4	92	3237	0	0	464	3716			472	32869	0	0
CD45	337	6773	0	0	1701	7775			2032	61578	0	0
CD56	12	363	0	0	60	416			61	3000	0	0
CD68	134	1037	0	0	676	1190			1900	6687	0	0
CD68_raw	247	933	0	0	1246	1071			2177	5710	0	0
CD69	233	1917	0	0	1176	2200			1195	18541	0	0
CD8	114	1343	0	0	575	1541			585	13245	0	0
area	0.20	0.87	0.00	0.00								

Table 6: AS1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	18	881	0	0	90	1728			92	4998	0	0
CD123	15	938	0	0	75	1840			76	5259	0	0
CD15	2	116	0	0	10	227			10	728	0	0
CD279	69	810	0	0	348	1589			354	5194	0	0
CD3	187	2536	0	0	944	4976			959	16568	0	0
CD4	92	1934	0	0	464	3795			472	12644	0	0
CD45	337	4323	0	0	1701	8483			2032	25755	0	0
CD56	12	235	0	0	60	461			61	1319	0	0
CD68	134	840	0	0	676	1648			1900	3976	0	0
CD68_raw	247	766	0	0	1246	1503			2177	3501	0	0
CD69	233	1163	0	0	1176	2282			1195	7034	0	0
CD8	114	808	0	0	575	1585			585	5034	0	0
area	0.20	0.51	0.00	0.00								

Table 7: AS1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1396	7069	0	0	2052	3499			3692	19029	0	0
CD123	605	3119	0	0	889	1543			1535	7817	0	0
CD279	624	3994	0	0	917	1976			2279	11773	0	0
CD3	1284	9955	0	0	1887	4927			3525	28840	0	0
CD4_2	1635	7638	0	0	2403	3780			6013	22417	0	0
CD4	2081	8021	0	0	3059	3970			6838	22527	0	0
CD8	819	3921	0	0	1204	1940			2304	10928	0	0
area	0.68	2.02	0.00	0.00								

Table 8: AS2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1396	5801	0	0	2052	4838			3692	11195	0	0
CD123	605	2630	0	0	889	2193			1535	4737	0	0
CD279	624	3200	0	0	917	2669			2279	6901	0	0
CD3	1284	7995	0	0	1887	6668			3525	17020	0	0
CD4_2	1635	6067	0	0	2403	5060			6013	12980	0	0
CD4	2081	6450	0	0	3059	5380			6838	13090	0	0
CD8	819	3167	0	0	1204	2641			2304	6252	0	0
area	0.68	1.20	0.00	0.00								

Table 9: AS2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD279	327	3620	0	0	569	1857			742	11869	0	0
CD3	578	10861	0	0	1005	5574			1261	36827	0	0
CD4	302	9296	0	0	525	4771			662	33106	0	0
CD56	77	1416	0	0	133	726			197	4611	0	0
CD8	538	4858	0	0	936	2493			1166	15415	0	0
area	0.57	1.95	0.00	0.00								

Table 10: AS3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD279	327	2753	0	0	569	2513			742	7182	0	0
CD3	578	7960	0	0	1005	7266			1261	20880	0	0
CD4	302	6627	0	0	525	6049			662	18438	0	0
CD56	77	925	0	0	133	844			197	1790	0	0
CD8	538	3577	0	0	936	3265			1166	8326	0	0
area	0.57	1.10	0.00	0.00								

Table 11: AS3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	631	7926	1505	4139	1443	6261	3886	5829	2907	47019	12230	38420
CD123	124	3900	407	1457	283	3080	1051	2052	590	23344	3150	10980
CD15	8	2460	934	403	18	1943	2412	567	43	15167	10037	2924
CD279	159	1622	59	1249	363	1281	152	1759	744	9212	469	11828
CD3	405	7078	277	5041	926	5591	715	7100	1787	41150	2077	48287
CD4	334	5588	216	3488	764	4414	557	4912	1496	33455	1501	35234
CD45	540	8700	995	5538	1235	6872	2569	7800	2398	51401	7396	52348
CD56	64	529	36	404	146	417	92	569	271	2650	253	3475
CD68	1576	3700	0	0	3605	2922	0	0	7300	16390	0	0
CD8	296	3668	210	2990	677	2897	542	4211	2096	20629	1683	28301
area	0.44	1.27	0.39	0.71								

Table 12: CF1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	631	4834	1505	2826	1443	7900	3886	7368	2907	23489	12230	18268
CD123	124	2464	407	1165	283	4026	1051	3037	590	12458	3150	6633
CD15	8	1680	934	353	18	2745	2412	920	43	9326	10037	2174
CD279	159	1134	59	905	363	1853	152	2359	744	5515	469	6876
CD3	405	4655	277	3728	926	7607	715	9720	1787	22743	2077	28866
CD4	334	3365	216	2385	764	5499	557	6218	1496	16515	1501	18569
CD45	540	5296	995	3915	1235	8655	2569	10207	2398	25411	7396	27922
CD56	64	373	36	308	146	609	92	803	271	1455	253	2051
CD68	1576	2886	0	0	3605	4716	0	0	7300	10181	0	0
CD8	296	2472	210	2217	677	4039	542	5780	2096	11536	1683	16863
area	0.44	0.61	0.39	0.38								

Table 13: CF1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1018	9592	752	1646	1861	6216	2566	5611	5118	44861	8087	39279
CD123	292	7591	234	783	534	4919	798	2669	1518	32922	2387	14618
CD279	189	2016	152	516	345	1306	518	1758	895	8951	1455	10018
CD3	330	5949	303	1632	603	3855	1034	5563	1558	27734	2906	32435
CD4	136	4208	148	1017	248	2727	505	3466	695	21421	1420	28231
CD8	263	4535	261	1632	481	2939	890	5563	1156	21495	2443	37062
area	0.55	1.54	0.29	0.29								

Table 14: CF2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1018	5869	752	1282	1861	7357	2566	6062	5118	20100	8087	14732
CD123	292	5287	234	692	534	6627	798	3272	1518	18736	2387	8062
CD279	189	1504	152	459	345	1885	518	2170	895	5682	1455	5670
CD3	330	4273	303	1465	603	5356	1034	6927	1558	16979	2906	18312
CD4	136	2685	148	810	248	3365	505	3830	695	11269	1420	11920
CD8	263	3123	261	1383	481	3915	890	6539	1156	12211	2443	18185
area	0.55	0.80	0.29	0.21								

Table 15: CF2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	698	3776	0	0	1901	1970			2786	10618	0	0
CD15	614	3611	0	0	1672	1884			2841	13052	0	0
CD279	85	2577	0	0	231	1344			348	9604	0	0
CD3	263	5572	0	0	716	2907			1055	22191	0	0
CD4	505	10867	0	0	1375	5670			2738	45293	0	0
CD45	1514	12844	0	0	4123	6701			6687	50171	0	0
CD56	1	279	0	0	2	145			3	1263	0	0
CD68	1576	3700	0	0	4292	1930			8352	11096	0	0
CD69	256	2961	0	0	697	1545			1371	11209	0	0
CD8	200	1943	0	0	544	1013			1037	7934	0	0
area	0.37	1.92	0.00	0.00								

Table 16: CF5_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	698	3522	0	0	1901	3809			2786	9354	0	0
CD15	614	3102	0	0	1672	3355			2841	10743	0	0
CD279	85	1959	0	0	231	2118			348	6536	0	0
CD3	263	3660	0	0	716	3958			1055	12386	0	0
CD4	505	6110	0	0	1375	6608			2738	21262	0	0
CD45	1514	7756	0	0	4123	8388			6687	24574	0	0
CD56	1	146	0	0	2	157			3	579	0	0
CD68	1576	2886	0	0	4292	3121			8352	6958	0	0
CD69	256	1843	0	0	697	1993			1371	5402	0	0
CD8	200	1123	0	0	544	1214			1037	3544	0	0
area	0.37	0.92	0.00	0.00								

Table 17: CF5_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	157	3201	0	0	306	1975			394	15202	0	0
CD123	13	1440	0	0	25	888			32	6833	0	0
CD15	1	88	0	0	1	54			2	377	0	0
CD279	141	841	0	0	275	518			403	3900	0	0
CD3	196	2168	0	0	382	1337			539	11192	0	0
CD4	397	4852	0	0	775	2994			1145	21972	0	0
CD45	529	7523	0	0	1033	4642			1466	35142	0	0
CD56	14	866	0	0	27	534			34	4280	0	0
CD68	581	2248	0	0	1134	1387			1789	9939	0	0
CD68_raw	263	1226	0	0	513	756			834	5560	0	0
CD69	194	1736	0	0	378	1071			576	8617	0	0
CD8	44	779	0	0	85	480			115	4134	0	0
area	0.51	1.62	0.00	0.00								

Table 18: CJ1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	157	2005	0	0	306	2243			394	6964	0	0
CD123	13	910	0	0	25	1018			32	3208	0	0
CD15	1	76	0	0	1	85			2	300	0	0
CD279	141	564	0	0	275	631			403	1985	0	0
CD3	196	1223	0	0	382	1368			539	4661	0	0
CD4	397	3153	0	0	775	3528			1145	10304	0	0
CD45	529	4751	0	0	1033	5316			1466	15955	0	0
CD56	14	405	0	0	27	453			34	1140	0	0
CD68	581	1492	0	0	1134	1669			1789	4779	0	0
CD68_raw	263	813	0	0	513	909			834	2766	0	0
CD69	194	1062	0	0	378	1188			576	3963	0	0
CD8	44	427	0	0	85	477			115	1696	0	0
area	0.51	0.89	0.00	0.00								

Table 19: CJ1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	323	4247	0	0	494	2246			662	16131	0	0
CD123	56	1739	0	0	85	919			110	6993	0	0
CD15	3	28	0	0	4	14			5	99	0	0
CD3	36	2392	0	0	55	1265			80	10534	0	0
CD4	60	2626	0	0	91	1388			143	11469	0	0
CD45	489	6667	0	0	749	3526			1035	26158	0	0
CD68	399	2450	0	0	611	1295			862	8827	0	0
CD8	46	805	0	0	70	425			115	3640	0	0
area	0.65	1.89	0.00	0.00								

Table 20: CJ2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	323	2937	0	0	494	2760			662	8314	0	0
CD123	56	1171	0	0	85	1100			110	3598	0	0
CD15	3	23	0	0	4	21			5	72	0	0
CD3	36	1508	0	0	55	1417			80	5276	0	0
CD4	60	1688	0	0	91	1586			143	5914	0	0
CD45	489	4462	0	0	749	4193			1035	12979	0	0
CD68	399	1820	0	0	611	1710			862	5090	0	0
CD8	46	474	0	0	70	445			115	1642	0	0
area	0.65	1.06	0.00	0.00								

Table 21: CJ2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	360	2203	10865	17752	3096	6651	8365	7851	7205	44740	43754	56722
CD123	308	1599	3062	14146	2649	4827	2357	6256	6258	28289	9729	44130
CD15	12	213	9301	3522	103	643	7161	1557	2180	3033	39371	8483
CD279	17	594	288	3275	146	1793	221	1448	318	12584	1015	11968
CD3	85	1535	140	6126	731	4634	107	2709	1741	30930	489	21917
CD4	233	2127	443	11429	2004	6421	341	5054	4824	42115	1359	40174
CD45	451	2935	12474	20869	3879	8861	9604	9230	11376	60192	50764	67113
CD56	7	134	3	612	60	404	2	270	131	3401	8	2272
CD68	141	673	1818	8087	1212	2031	1399	3576	3040	13207	6153	25334
CD69	19	253	31	2283	163	763	23	1009	356	5665	141	8652
CD8	37	634	75	2692	318	1914	57	1190	745	14085	254	9868
area	0.12	0.33	1.30	2.26								

Table 22: DD1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	360	1551	10865	11760	3096	9814	8365	10881	7205	25974	43754	31464
CD123	308	1463	3062	10459	2649	9257	2357	9677	6258	24199	9729	28719
CD15	12	211	9301	3230	103	1335	7161	2988	2180	2974	39371	7291
CD279	17	512	288	2289	146	3239	221	2118	318	10361	1015	7856
CD3	85	1308	140	4119	731	8276	107	3811	1741	24708	489	13512
CD4	233	1648	443	7342	2004	10427	341	6793	4824	28552	1359	22956
CD45	451	2153	12474	13998	3879	13623	9604	12952	11376	37679	50764	38183
CD56	7	101	3	381	60	639	2	352	131	2453	8	1302
CD68	141	497	1818	5491	1212	3144	1399	5080	3040	8057	6153	14408
CD69	19	216	31	1188	163	1366	23	1099	356	4682	141	4076
CD8	37	503	75	1685	318	3182	57	1559	745	10533	254	5641
area	0.12	0.16	1.30	1.08								

Table 23: DD1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	875	1740	1003	7548	3036	2783	1294	2853	7053	22057	1576	20163
CD123	266	743	759	5397	923	1188	979	2040	2279	6978	1460	8894
CD15	12	395	2932	3889	41	631	3784	1470	139	3852	16106	3487
CD279	27	40	37	128	93	63	47	48	226	421	65	280
CD3	293	2213	449	6549	1016	3540	579	2475	2662	22916	844	13175
CD4	241	1276	313	4330	836	2041	404	1636	2055	13312	529	9541
CD45	1148	4987	3613	22440	3984	7979	4664	8482	10267	54768	9448	48032
CD56	30	88	72	449	104	140	92	169	264	899	112	1131
CD68	162	249	413	1451	562	398	533	548	1371	2149	1218	2825
CD69	530	1530	578	5345	1839	2447	746	2020	4454	17518	898	10898
CD8	124	615	129	1683	430	983	166	636	1051	5981	208	3520
area	0.29	0.63	0.77	2.65								

Table 24: DD2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	875	831	1003	5243	3036	3224	1294	3477	7053	10538	1576	9986
CD123	266	554	759	5097	923	2149	979	3381	2279	4556	1460	7690
CD15	12	381	2932	3850	41	1478	3784	2553	139	3665	16106	3328
CD279	27	33	37	111	93	128	47	73	226	330	65	213
CD3	293	1634	449	5844	1016	6341	579	3876	2662	15079	844	10082
CD4	241	891	313	3654	836	3457	404	2423	2055	8220	529	6654
CD45	1148	3160	3613	17682	3984	12262	4664	11729	10267	30284	9448	27219
CD56	30	44	72	343	104	170	92	227	264	426	112	705
CD68	162	184	413	1207	562	714	533	800	1371	1318	1218	1791
CD69	530	1011	578	4772	1839	3923	746	3165	4454	10386	898	8520
CD8	124	469	129	1473	430	1820	166	977	1051	3920	208	2631
area	0.29	0.26	0.77	1.51								

Table 25: DD2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	478	2859	10260	17020	2745	6305	8695	6913	5194	42397	47109	44446
CD123	458	2163	3003	13393	2630	4770	2545	5440	4761	27254	7639	29442
CD15	46	529	8776	3516	264	1166	7437	1428	1073	6368	41005	5953
CD279	22	258	186	1580	126	568	157	641	237	4698	449	4312
CD3	97	1086	355	3700	557	2395	300	1502	1080	17451	799	10727
CD4	372	2498	598	12097	2136	5509	506	4913	3918	36737	1312	35212
CD45	509	3197	11142	17714	2923	7050	9442	7195	5589	46301	52430	46066
CD56	18	159	41	669	103	350	34	271	208	2951	84	2102
CD68	344	1487	2300	10029	1975	3279	1949	4073	5034	22858	7049	26343
CD69	2	63	96	388	11	138	81	157	28	1208	198	1215
CD8	61	722	211	2741	350	1592	178	1113	691	12632	514	8445
area	0.17	0.45	1.18	2.46								

Table 26: DD3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	478	1912	10260	12619	2745	10028	8695	9777	5194	24443	47109	25741
CD123	458	1902	3003	12031	2630	9975	2545	9321	4761	22343	7639	23875
CD15	46	524	8776	3470	264	2748	7437	2688	1073	6274	41005	5765
CD279	22	228	186	1228	126	1195	157	951	237	4132	449	2841
CD3	97	935	355	2563	557	4903	300	1985	1080	14626	799	5855
CD4	372	1870	598	8248	2136	9807	506	6390	3918	24933	1312	18841
CD45	509	2356	11142	13171	2923	12356	9442	10205	5589	30388	52430	26799
CD56	18	101	41	413	103	529	34	320	208	1847	84	1008
CD68	344	922	2300	7518	1975	4835	1949	5825	5034	12131	7049	15733
CD69	2	53	96	288	11	277	81	223	28	1028	198	822
CD8	61	551	211	1677	350	2889	178	1299	691	9424	514	3865
area	0.17	0.19	1.18	1.29								

Table 27: DD3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	56	5795	0	0	177	3802			257	32673	0	0
CD123	47	3656	0	0	148	2398			199	19758	0	0
CD15	9	178	0	0	28	116			36	793	0	0
CD279	45	203	0	0	142	133			181	1074	0	0
CD3	143	8872	0	0	452	5820			575	50879	0	0
CD4	221	10807	0	0	699	7090			1050	59526	0	0
CD45	391	14245	0	0	1236	9346			1615	77705	0	0
CD56	5	825	0	0	15	541			30	4592	0	0
CD68	121	4464	0	0	382	2928			551	24026	0	0
CD8	62	3844	0	0	196	2522			368	22116	0	0
area	0.32	1.52	0.00	0.00								

Table 28: EB_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	56	2940	0	0	177	4463			257	16057	0	0
CD123	47	2025	0	0	148	3074			199	10257	0	0
CD15	9	129	0	0	28	195			36	507	0	0
CD279	45	121	0	0	142	183			181	599	0	0
CD3	143	4500	0	0	452	6832			575	25489	0	0
CD4	221	5758	0	0	699	8742			1050	30173	0	0
CD45	391	7493	0	0	1236	11376			1615	38563	0	0
CD56	5	426	0	0	15	646			30	2273	0	0
CD68	121	2445	0	0	382	3712			551	12303	0	0
CD8	62	1852	0	0	196	2811			368	10613	0	0
area	0.32	0.66	0.00	0.00								

Table 29: EB_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	255	3684	0	0	309	3277			585	18349	0	0
CD123	22	435	0	0	26	386			44	2053	0	0
CD15	74	1738	0	0	89	1546			1646	13393	0	0
CD279	22	97	0	0	26	86			63	310	0	0
CD3	26	391	0	0	31	347			50	1321	0	0
CD4	69	1659	0	0	83	1475			189	9044	0	0
CD56	6	148	0	0	7	131			11	643	0	0
CD68	1321	1568	0	0	1601	1394			2983	10585	0	0
CD68_raw	1701	2379	0	0	2062	2116			3900	14474	0	0
CD69	33	165	0	0	40	146			64	390	0	0
CD8	24	110	0	0	29	97			46	246	0	0
area	0.82	1.12	0.00	0.00								

Table 30: JK1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	255	3235	0	0	309	3340			585	10971	0	0
CD123	22	367	0	0	26	378			44	796	0	0
CD15	74	1401	0	0	89	1446			1646	6714	0	0
CD279	22	93	0	0	26	96			63	212	0	0
CD3	26	370	0	0	31	382			50	985	0	0
CD4	69	1403	0	0	83	1448			189	5058	0	0
CD56	6	132	0	0	7	136			11	330	0	0
CD68	1321	1151	0	0	1601	1188			2983	4141	0	0
CD68_raw	1701	1851	0	0	2062	1911			3900	6145	0	0
CD69	33	156	0	0	40	161			64	279	0	0
CD8	24	110	0	0	29	113			46	246	0	0
area	0.82	0.97	0.00	0.00								

Table 31: JK1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	18	417	0	0	23	373			32	2049	0	0
CD4	86	1748	0	0	112	1563			216	7623	0	0
CD8	13	97	0	0	17	86			27	269	0	0
area	0.76	1.12	0.00	0.00								

Table 32: JK2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	18	334	0	0	23	390			32	1129	0	0
CD4	86	1484	0	0	112	1735			216	4426	0	0
CD8	13	95	0	0	17	111			27	230	0	0
area	0.76	0.86	0.00	0.00								

Table 33: JK2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	17	459	0	0	23	462			36	3431	0	0
CD15	7	717	0	0	9	722			19	6137	0	0
CD279	17	146	0	0	23	147			35	766	0	0
CD3	20	251	0	0	27	253			42	1295	0	0
CD4	68	1422	0	0	94	1433			154	5903	0	0
CD68	756	1795	0	0	1045	1809			1882	8426	0	0
CD68_raw	1263	1499	0	0	1745	1511			3015	7592	0	0
CD69	22	158	0	0	30	159			91	1070	0	0
CD8	15	56	0	0	20	56			31	169	0	0
area	0.72	0.99	0.00	0.00								

Table 34: JK3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	17	372	0	0	23	461			36	1025	0	0
CD15	7	499	0	0	9	618			19	2427	0	0
CD279	17	127	0	0	23	157			35	328	0	0
CD3	20	224	0	0	27	277			42	781	0	0
CD4	68	1316	0	0	94	1631			154	4090	0	0
CD68	756	1595	0	0	1045	1977			1882	4858	0	0
CD68_raw	1263	1311	0	0	1745	1625			3015	4020	0	0
CD69	22	133	0	0	30	164			91	290	0	0
CD8	15	52	0	0	20	64			31	104	0	0
area	0.72	0.81	0.00	0.00								

Table 35: JK3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	929	5747	1199	7120	1912	5058	4988	8169	2992	22825	7293	46629
CD123	376	3507	649	5802	773	3086	2700	6656	1161	11487	3566	34103
CD15	6	122	326	345	12	107	1356	395	22	398	2312	1437
CD279	182	2440	62	1946	374	2147	257	2232	568	8567	337	14251
CD3	336	6967	155	4928	691	6131	644	5654	1067	32393	916	43238
CD4	621	7178	448	5687	1278	6317	1863	6525	1983	32235	2730	45746
CD45	1238	11869	1638	10638	2548	10446	6814	12205	4000	61369	11128	84620
CD56	152	670	44	495	312	589	183	567	465	2431	239	3978
CD68	623	2813	1122	4096	1282	2475	4667	4699	2008	9515	7316	21477
CD68_raw	860	2235	931	3089	1770	1967	3873	3544	2799	7405	6400	16275
CD69	731	7062	898	7673	1504	6215	3736	8803	2271	26876	5068	51526
CD8	176	2236	62	1565	362	1967	257	1795	550	9114	349	13401
area	0.49	1.14	0.24	0.87								

Table 36: JM1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	929	5124	1199	5925	1912	7690	4988	11805	2992	16621	7293	32173
CD123	376	3271	649	5209	773	4909	2700	10378	1161	9037	3566	26921
CD15	6	103	326	333	12	154	1356	663	22	185	2312	1276
CD279	182	2342	62	1678	374	3514	257	3343	568	7547	337	11084
CD3	336	6090	155	3206	691	9139	644	6387	1067	23153	916	21119
CD4	621	6232	448	4008	1278	9352	1863	7985	1983	22375	2730	24358
CD45	1238	9412	1638	7482	2548	14125	6814	14907	4000	35821	11128	44025
CD56	152	615	44	344	312	922	183	685	465	1870	239	2128
CD68	623	2592	1122	3759	1282	3890	4667	7489	2008	7275	7316	17478
CD68_raw	860	2071	931	2864	1770	3108	3873	5706	2799	5743	6400	13612
CD69	731	6708	898	6397	1504	10067	3736	12745	2271	23302	5068	36346
CD8	176	2039	62	994	362	3060	257	1980	550	7028	349	6321
area	0.49	0.67	0.24	0.50								

Table 37: JM1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1561	4294	0	0	2517	4465			4654	14702	0	0
CD123	298	2103	0	0	480	2187			678	7352	0	0
CD15	12	76	0	0	19	79			228	219	0	0
CD279	161	511	0	0	259	531			362	1322	0	0
CD3	403	4870	0	0	649	5064			912	13728	0	0
CD4	656	5371	0	0	1058	5585			1510	20365	0	0
CD45	798	4536	0	0	1287	4717			1918	12324	0	0
CD56	188	678	0	0	303	705			425	1876	0	0
CD68	848	1857	0	0	1367	1931			2693	5393	0	0
CD68_raw	2839	2193	0	0	4578	2280			8979	6295	0	0
CD69	791	6050	0	0	1275	6291			1766	17855	0	0
CD8	211	2087	0	0	340	2170			460	5812	0	0
area	0.62	0.96	0.00	0.00								

Table 38: JM2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1561	4161	0	0	2517	5588			4654	11580	0	0
CD123	298	2055	0	0	480	2760			678	6081	0	0
CD15	12	71	0	0	19	95			228	102	0	0
CD279	161	509	0	0	259	683			362	1264	0	0
CD3	403	4813	0	0	649	6464			912	12486	0	0
CD4	656	5185	0	0	1058	6964			1510	15397	0	0
CD45	798	4499	0	0	1287	6042			1918	11184	0	0
CD56	188	666	0	0	303	894			425	1620	0	0
CD68	848	1817	0	0	1367	2440			2693	4751	0	0
CD68_raw	2839	2159	0	0	4578	2899			8979	5635	0	0
CD69	791	5997	0	0	1275	8054			1766	16458	0	0
CD8	211	2063	0	0	340	2770			460	4972	0	0
area	0.62	0.74	0.00	0.00								

Table 39: JM2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	233	4303	0	0	449	5757			692	13715	0	0
CD8	170	1750	0	0	328	2341			511	5158	0	0
area	0.52	0.75	0.00	0.00								

Table 40: JM3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	233	4264	0	0	449	6414			692	12597	0	0
CD8	170	1734	0	0	328	2608			511	4675	0	0
area	0.52	0.66	0.00	0.00								

Table 41: JM3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	24	1182	0	0	107	781			107	3843	0	0
CD123	9	896	0	0	40	592			40	3075	0	0
CD15	4	218	0	0	17	144			17	943	0	0
CD279	15	149	0	0	66	98			66	522	0	0
CD3	92	1195	0	0	410	790			410	4067	0	0
CD4	72	1189	0	0	321	786			321	4144	0	0
CD45	161	3393	0	0	718	2243			718	12420	0	0
CD56	8	174	0	0	35	115			35	467	0	0
CD68	243	923	0	0	1084	610			1080	3126	0	0
CD68_raw	558	956	0	0	2489	632			2490	3296	0	0
CD8	54	518	0	0	240	342			241	2000	0	0
area	0.22	1.51	0.00	0.00								

Table 42: JT_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	24	1125	0	0	107	1311			107	3425	0	0
CD123	9	824	0	0	40	960			40	2522	0	0
CD15	4	179	0	0	17	208			17	661	0	0
CD279	15	134	0	0	66	156			66	400	0	0
CD3	92	1139	0	0	410	1328			410	3629	0	0
CD4	72	1105	0	0	321	1288			321	3460	0	0
CD45	161	3082	0	0	718	3594			718	9948	0	0
CD56	8	155	0	0	35	180			35	317	0	0
CD68	243	847	0	0	1084	987			1080	2536	0	0
CD68_raw	558	868	0	0	2489	1012			2490	2571	0	0
CD8	54	462	0	0	240	538			241	1548	0	0
area	0.22	0.86	0.00	0.00								

Table 43: JT_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	41	2501	0	0	118	4100			212	20605	0	0
CD123	20	1599	0	0	58	2621			74	11693	0	0
CD15	5	176	0	0	14	288			48	1762	0	0
CD279	89	440	0	0	258	721			381	2401	0	0
CD3	540	3111	0	0	1566	5100			2094	20040	0	0
CD4	209	2014	0	0	606	3302			829	14072	0	0
CD45	794	5665	0	0	2302	9288			3180	39615	0	0
CD56	48	144	0	0	139	236			179	689	0	0
CD68	233	862	0	0	675	1413			968	4728	0	0
CD69	392	2232	0	0	1136	3659			1512	16013	0	0
CD8	223	993	0	0	646	1628			832	5963	0	0
area	0.34	0.61	0.00	0.00								

Table 44: LD_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	41	2167	0	0	118	4715			212	16310	0	0
CD123	20	1528	0	0	58	3324			74	10819	0	0
CD15	5	165	0	0	14	359			48	1041	0	0
CD279	89	433	0	0	258	942			381	2324	0	0
CD3	540	2992	0	0	1566	6510			2094	18412	0	0
CD4	209	1868	0	0	606	4064			829	11932	0	0
CD45	794	5078	0	0	2302	11048			3180	31704	0	0
CD56	48	143	0	0	139	311			179	670	0	0
CD68	233	820	0	0	675	1784			968	4240	0	0
CD69	392	2156	0	0	1136	4691			1512	14526	0	0
CD8	223	948	0	0	646	2062			832	5295	0	0
area	0.34	0.46	0.00	0.00								

Table 45: LD_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1074	5057	942	5215	2325	3056	4586	5237	3520	19296	7543	37327
CD123	132	916	613	1498	285	553	2984	1504	405	2448	4437	5717
CD15	6	101	14	49	12	61	68	49	23	320	143	301
CD279	67	339	13	257	145	204	63	258	206	813	84	2144
CD3	205	2000	83	1327	443	1208	404	1332	610	6164	554	10356
CD4	303	2587	95	2189	656	1563	462	2198	905	8683	672	17517
CD45	967	6473	798	5806	2093	3912	3885	5830	3051	24134	6186	43817
CD56	34	247	6	242	73	149	29	243	110	703	39	2022
CD69	169	1166	73	1225	365	704	355	1230	560	3303	496	9320
CD8	178	1663	69	1223	385	1005	335	1228	553	5818	463	9485
area	0.46	1.65	0.21	1.00								

Table 46: LG1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1074	3962	942	3627	2325	4918	4586	8633	3520	12879	7543	23282
CD123	132	828	613	1376	285	1027	2984	3275	405	1943	4437	4649
CD15	6	87	14	32	12	107	68	76	23	233	143	148
CD279	67	319	13	195	145	395	63	464	206	695	84	1589
CD3	205	1731	83	946	443	2148	404	2251	610	4652	554	6978
CD4	303	2272	95	1509	656	2820	462	3591	905	6793	672	11475
CD45	967	5210	798	3952	2093	6467	3885	9407	3051	16698	6186	27396
CD56	34	224	6	161	73	278	29	383	110	560	39	1300
CD69	169	1109	73	1001	365	1376	355	2382	560	2963	496	7331
CD8	178	1348	69	892	385	1673	335	2123	553	4038	463	6569
area	0.46	0.81	0.21	0.42								

Table 47: LG1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	93	5966	0	0	274	2430			348	18757	0	0
CD123	5	1047	0	0	14	426			15	3282	0	0
CD15	1	1619	0	0	2	659			3	5904	0	0
CD279	43	537	0	0	126	218			132	1727	0	0
CD3	58	4379	0	0	170	1783			178	15120	0	0
CD4	92	5078	0	0	271	2068			348	16951	0	0
CD45	194	9993	0	0	571	4070			659	32381	0	0
CD56	8	874	0	0	23	355			24	2897	0	0
CD68	962	3235	0	0	2836	1317			3281	8861	0	0
CD68_raw	967	1731	0	0	2850	705			3041	4593	0	0
CD69	98	2628	0	0	288	1070			301	8514	0	0
CD8	40	2504	0	0	117	1019			123	8530	0	0
area	0.34	2.46	0.00	0.00								

Table 48: LG2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	93	3583	0	0	274	3081			348	9576	0	0
CD123	5	691	0	0	14	594			15	1910	0	0
CD15	1	578	0	0	2	497			3	1867	0	0
CD279	43	349	0	0	126	300			132	1001	0	0
CD3	58	2497	0	0	170	2147			178	7872	0	0
CD4	92	3054	0	0	271	2626			348	9146	0	0
CD45	194	5971	0	0	571	5135			659	16893	0	0
CD56	8	420	0	0	23	361			24	1163	0	0
CD68	962	2212	0	0	2836	1902			3281	4922	0	0
CD68_raw	967	1200	0	0	2850	1032			3041	2552	0	0
CD69	98	1817	0	0	288	1562			301	5400	0	0
CD8	40	1456	0	0	117	1252			123	4496	0	0
area	0.34	1.16	0.00	0.00								

Table 49: LG2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	287	2999	149	2526	632	1716	472	2489	886	9631	1032	17512
CD4	403	3396	241	3053	887	1944	763	3008	1227	12074	1447	21021
CD68	828	1401	1139	1512	1824	801	3608	1489	2943	4648	7797	6171
CD68_raw	1527	2002	1657	2178	3364	1146	5250	2146	5489	6722	11556	9982
CD8	187	1791	97	1647	411	1025	307	1622	584	5793	735	12035
area	0.45	1.75	0.32	1.01								

Table 50: LG3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	287	2600	149	1901	632	3163	472	4112	886	7472	1032	11855
CD4	403	2796	241	2200	887	3401	763	4758	1227	8857	1447	13302
CD68	828	1113	1139	1276	1824	1354	3608	2760	2943	3110	7797	4035
CD68_raw	1527	1565	1657	1752	3364	1904	5250	3789	5489	4391	11556	6128
CD8	187	1558	97	1199	411	1895	307	2593	584	4514	735	7980
area	0.45	0.82	0.32	0.46								

Table 51: LG3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	298	7681	0	0	745	4314			871	30612	0	0
CD123	18	1632	0	0	45	916			49	5907	0	0
CD15	65	517	0	0	162	290			179	1292	0	0
CD279	39	177	0	0	97	99			107	549	0	0
CD3	225	9871	0	0	563	5544			646	39738	0	0
CD4	354	16036	0	0	885	9006			1003	65640	0	0
CD45	482	18467	0	0	1206	10372			1427	77612	0	0
CD56	45	926	0	0	112	520			124	3538	0	0
CD68	88	1563	0	0	220	877			243	5272	0	0
CD69	77	2900	0	0	192	1628			213	12362	0	0
CD8	125	4096	0	0	312	2300			369	17290	0	0
area	0.40	1.78	0.00	0.00								

Table 52: LM_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	298	5322	0	0	745	5360			871	16006	0	0
CD123	18	1229	0	0	45	1237			49	3442	0	0
CD15	65	475	0	0	162	478			179	1034	0	0
CD279	39	134	0	0	97	134			107	298	0	0
CD3	225	6814	0	0	563	6862			646	21252	0	0
CD4	354	10892	0	0	885	10970			1003	33959	0	0
CD45	482	12185	0	0	1206	12272			1427	38589	0	0
CD56	45	658	0	0	112	662			124	1871	0	0
CD68	88	1212	0	0	220	1220			243	3178	0	0
CD69	77	1976	0	0	192	1990			213	6831	0	0
CD8	125	2635	0	0	312	2653			369	8447	0	0
area	0.40	0.99	0.00	0.00								

Table 53: LM_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1402	7533	0	0	2030	3501			3949	21233	0	0
CD123	119	1401	0	0	172	651			334	4008	0	0
CD15	16	49	0	0	23	22			86	133	0	0
CD279	496	4097	0	0	718	1904			1321	14529	0	0
CD3	692	10338	0	0	1002	4804			1633	35819	0	0
CD4	813	10287	0	0	1177	4781			2016	33067	0	0
CD45	1692	14059	0	0	2450	6534			4327	45292	0	0
CD56	133	1209	0	0	192	561			312	4186	0	0
CD68	908	1720	0	0	1315	799			2368	4082	0	0
CD68_raw	1423	2347	0	0	2060	1090			3706	5522	0	0
CD69	1009	7850	0	0	1461	3648			2446	25027	0	0
CD8	291	3429	0	0	421	1593			692	11744	0	0
area	0.69	2.15	0.00	0.00								

Table 54: LS1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1402	5431	0	0	2030	4548			3949	10483	0	0
CD123	119	978	0	0	172	819			334	1867	0	0
CD15	16	44	0	0	23	36			86	106	0	0
CD279	496	2310	0	0	718	1934			1321	5329	0	0
CD3	692	6396	0	0	1002	5357			1633	15492	0	0
CD4	813	6816	0	0	1177	5709			2016	15273	0	0
CD45	1692	9219	0	0	2450	7721			4327	20411	0	0
CD56	133	636	0	0	192	532			312	1180	0	0
CD68	908	1309	0	0	1315	1096			2368	1959	0	0
CD68_raw	1423	1807	0	0	2060	1513			3706	2729	0	0
CD69	1009	5192	0	0	1461	4348			2446	11494	0	0
CD8	291	2117	0	0	421	1773			692	4992	0	0
area	0.69	1.19	0.00	0.00								

Table 55: LS1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1567	7637	0	0	1490	3340			3296	21412	0	0
CD123	477	3198	0	0	453	1399			1030	7797	0	0
CD15	6	30	0	0	5	13			12	75	0	0
CD279	342	1547	0	0	325	676			603	4389	0	0
CD3	980	11077	0	0	932	4845			1687	33467	0	0
CD4	1229	12721	0	0	1168	5565			2143	37434	0	0
CD45	1941	15205	0	0	1845	6651			3742	45520	0	0
CD56	72	644	0	0	68	281			119	2287	0	0
CD68	1610	3072	0	0	1531	1343			3155	7119	0	0
CD68_raw	3474	4839	0	0	3303	2116			7336	11341	0	0
CD69	652	5312	0	0	620	2323			1119	15818	0	0
CD8	302	2926	0	0	287	1280			505	8709	0	0
area	1.05	2.29	0.00	0.00								

Table 56: LS2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1567	5051	0	0	1490	4010			3296	8940	0	0
CD123	477	2355	0	0	453	1870			1030	3751	0	0
CD15	6	27	0	0	5	21			12	60	0	0
CD279	342	1072	0	0	325	851			603	2072	0	0
CD3	980	7326	0	0	932	5817			1687	15380	0	0
CD4	1229	8433	0	0	1168	6696			2143	16836	0	0
CD45	1941	9942	0	0	1845	7894			3742	20187	0	0
CD56	72	275	0	0	68	218			119	447	0	0
CD68	1610	2329	0	0	1531	1849			3155	3559	0	0
CD68_raw	3474	3626	0	0	3303	2879			7336	5521	0	0
CD69	652	3576	0	0	620	2839			1119	7449	0	0
CD8	302	1897	0	0	287	1506			505	3725	0	0
area	1.05	1.26	0.00	0.00								

Table 57: LS2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	643	11896	0	0	646	5140			1152	36142	0	0
CD4	797	13016	0	0	800	5624			1507	37868	0	0
CD68	2794	3479	0	0	2807	1503			5603	7474	0	0
CD68_raw	4129	4884	0	0	4149	2110			8835	10839	0	0
CD8	319	3956	0	0	320	1709			540	11834	0	0
area	1.00	2.31	0.00	0.00								

Table 58: LS3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	643	7999	0	0	646	6209			1152	17397	0	0
CD4	797	8700	0	0	800	6753			1507	16986	0	0
CD68	2794	2721	0	0	2807	2112			5603	3771	0	0
CD68_raw	4129	3720	0	0	4149	2887			8835	5165	0	0
CD8	319	2507	0	0	320	1946			540	4829	0	0
area	1.00	1.29	0.00	0.00								

Table 59: LS3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	84	2383	3133	4770	909	3323	3948	4900	1129	19527	13606	17775
CD123	37	1361	1582	2848	400	1898	1993	2925	425	10599	4810	8845
CD15	18	475	1775	1494	194	662	2237	1534	206	3411	15051	5021
CD279	43	791	545	1384	465	1103	686	1421	493	6510	1669	3561
CD3	69	1063	717	1686	746	1482	903	1732	792	8403	1889	5629
CD4	30	952	421	1960	324	1327	530	2013	344	8031	1068	5633
CD45	199	3631	4090	7433	2154	5064	5155	7636	2285	29107	17812	28517
CD56	6	262	41	380	64	365	51	390	68	2310	91	1337
CD8	36	461	315	965	389	642	397	991	413	3658	817	2221
area	0.09	0.72	0.79	0.97								

Table 60: SK1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	84	2077	3133	4329	909	5258	3948	6156	1129	15679	13606	8636
CD123	37	1278	1582	2642	400	3235	1993	3757	425	9527	4810	4170
CD15	18	441	1775	1418	194	1116	2237	2016	206	3005	15051	3621
CD279	43	773	545	1311	465	1957	686	1864	493	6300	1669	1761
CD3	69	1046	717	1548	746	2648	903	2201	792	8199	1889	2055
CD4	30	928	421	1850	324	2349	530	2631	344	7739	1068	2801
CD45	199	3257	4090	6726	2154	8245	5155	9565	2285	24147	17812	13674
CD56	6	254	41	350	64	643	51	497	68	2216	91	700
CD8	36	450	315	919	389	1139	397	1306	413	3513	817	1102
area	0.09	0.39	0.79	0.70								

Table 61: SK1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1024	7035	0	0	1491	3216			3213	21119	0	0
CD123	96	945	0	0	139	432			387	2447	0	0
CD15	344	1075	0	0	500	491			1185	2607	0	0
CD279	136	827	0	0	198	378			406	2194	0	0
CD3	343	4600	0	0	499	2103			1266	13609	0	0
CD4	580	5479	0	0	844	2505			1811	15880	0	0
CD45	758	7130	0	0	1103	3259			2406	21562	0	0
CD56	32	281	0	0	46	128			136	782	0	0
CD68	551	1528	0	0	802	698			1764	3957	0	0
CD69	744	3693	0	0	1083	1688			2522	9820	0	0
CD8	146	914	0	0	212	417			482	2578	0	0
area	0.69	2.19	0.00	0.00								

Table 62: SK2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1024	5210	0	0	1491	4700			3213	12884	0	0
CD123	96	819	0	0	139	738			387	1906	0	0
CD15	344	834	0	0	500	752			1185	1483	0	0
CD279	136	784	0	0	198	707			406	2012	0	0
CD3	343	4049	0	0	499	3653			1266	11207	0	0
CD4	580	4636	0	0	844	4183			1811	12133	0	0
CD45	758	5575	0	0	1103	5030			2406	14607	0	0
CD56	32	221	0	0	46	199			136	516	0	0
CD68	551	1268	0	0	802	1144			1764	2802	0	0
CD69	744	3230	0	0	1083	2914			2522	7877	0	0
CD8	146	825	0	0	212	744			482	2191	0	0
area	0.69	1.11	0.00	0.00								

Table 63: SK2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	444	4568	0	0	740	1967			891	12313	0	0
CD123	135	2885	0	0	225	1242			268	8055	0	0
CD279	34	124	0	0	56	53			67	314	0	0
CD3	258	5477	0	0	430	2359			599	16013	0	0
CD4	355	5541	0	0	592	2386			790	15836	0	0
CD68	1047	1423	0	0	1746	613			3140	3008	0	0
CD68_raw	1946	3284	0	0	3245	1414			5326	7196	0	0
CD8	123	1305	0	0	205	562			279	3601	0	0
area	0.60	2.32	0.00	0.00								

Table 64: SK3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	444	3423	0	0	740	2953			891	7423	0	0
CD123	135	2425	0	0	225	2092			268	6160	0	0
CD279	34	113	0	0	56	97			67	267	0	0
CD3	258	4809	0	0	430	4149			599	13206	0	0
CD4	355	4797	0	0	592	4139			790	12669	0	0
CD68	1047	1314	0	0	1746	1133			3140	2547	0	0
CD68_raw	1946	2907	0	0	3245	2508			5326	5611	0	0
CD8	123	1212	0	0	205	1045			279	3207	0	0
area	0.60	1.16	0.00	0.00								

Table 65: SK3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.